MFDStudio - Detailed User Guide

design in `mfd_editor`, C++17 API generation, live client integration, offscreen runtime embedding, framebuffer plugin, and launch scripts

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Scope: editor, architecture, generator, client API, offscreen runtime embedding, framebuffer plugin, launch scripts
Target audience: MFD authors, C++ integrators, embedding application teams, and runtime teams

Editor · ch. 1-4	API · ch. 5-6	Plugin · ch. 7

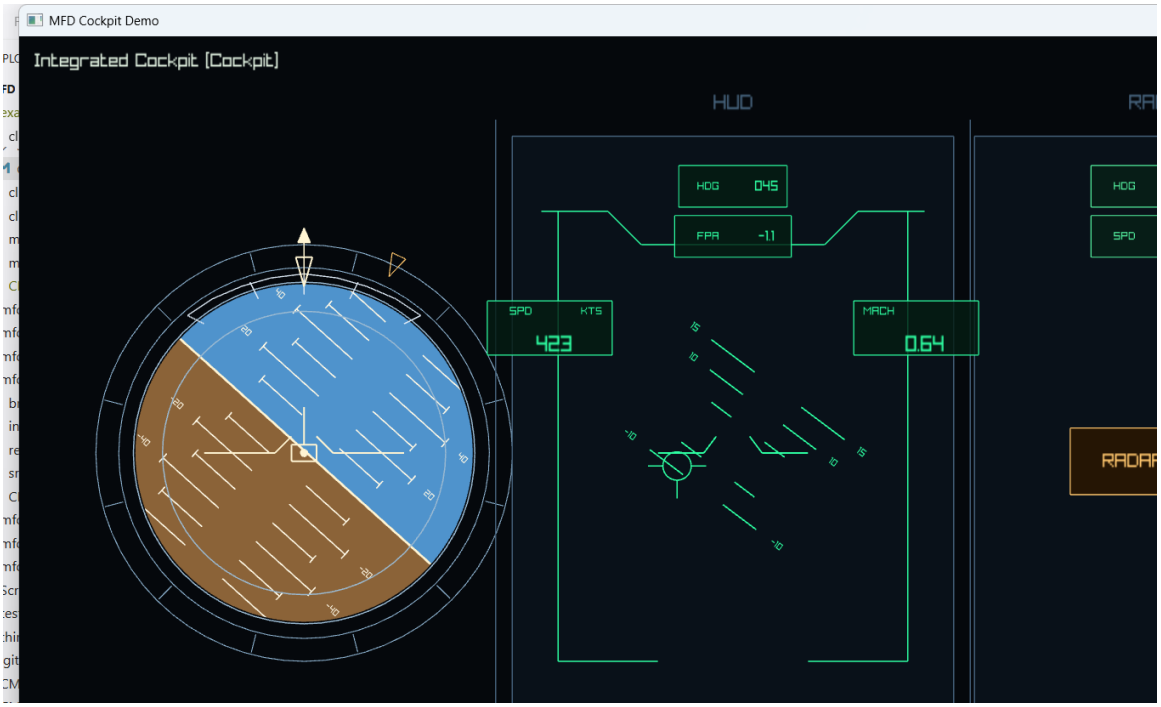
Target document:

- MFD page and window authors
- C++ integrators who must control a window live
- teams who must embed the runtime offscreen inside their own application
- teams who want to capture the framebuffer via a DLL plugin

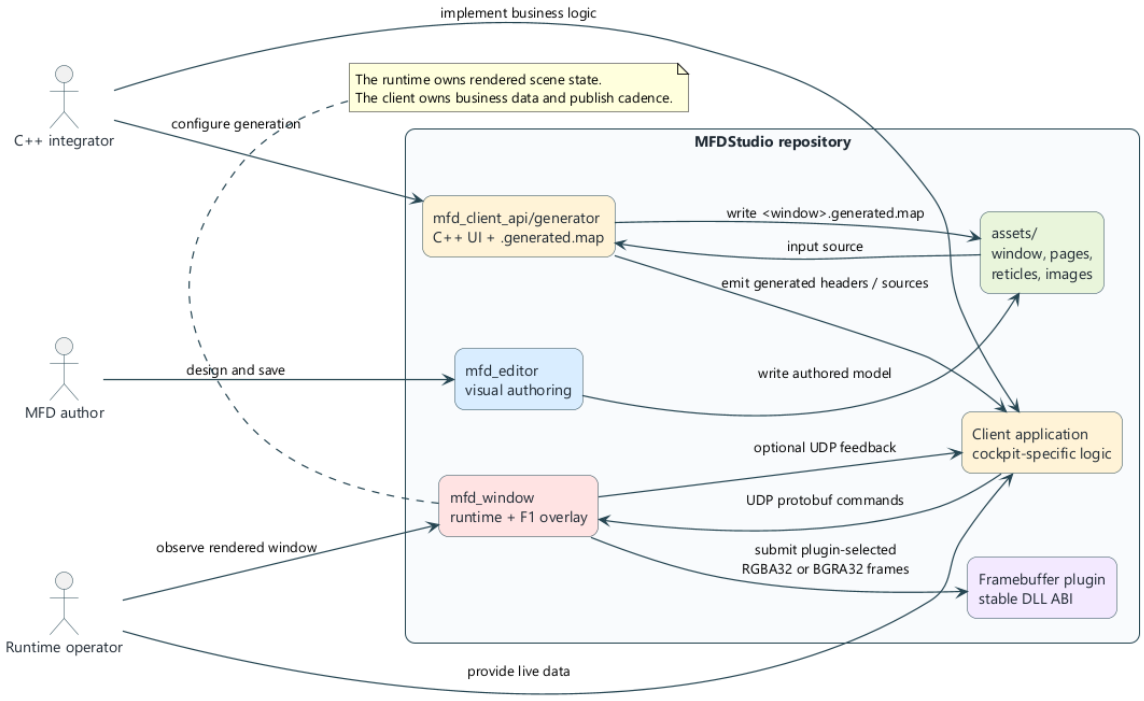
Table of Contents

The table of contents will be updated by Word.

1. What is MFDStudio



Cockpit runtime screenshot



MFDStudio Overview

MFDStudio is a C++17 / CMake toolbox for designing, executing and controlling MFD type 2D windows from JSON files. The repository voluntarily separates three responsibilities:

- the content author defines versioned assets: window, pages, reticles, images, fonts
- the runtime `mfd_window` loads these assets and renders the active page

- the external client sends live commands via UDP to modify the runtime state without rewriting the assets

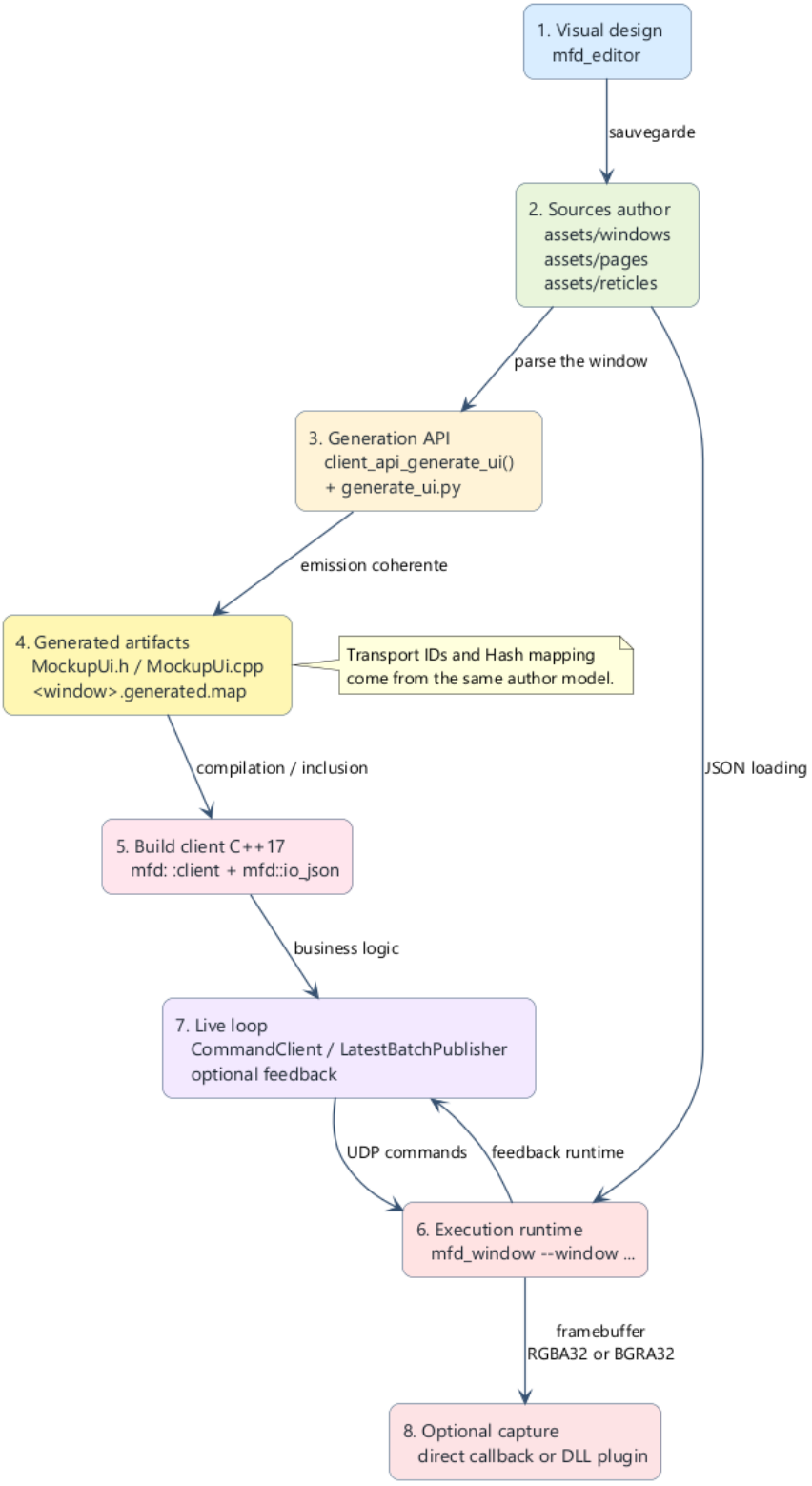
The historical technical prefix of the repository remains `mfd` in namespaces, CMake targets, folders, and public APIs. You must therefore distinguish:

- the product name: `MFDStudio`
- the technical prefix: `mfd`

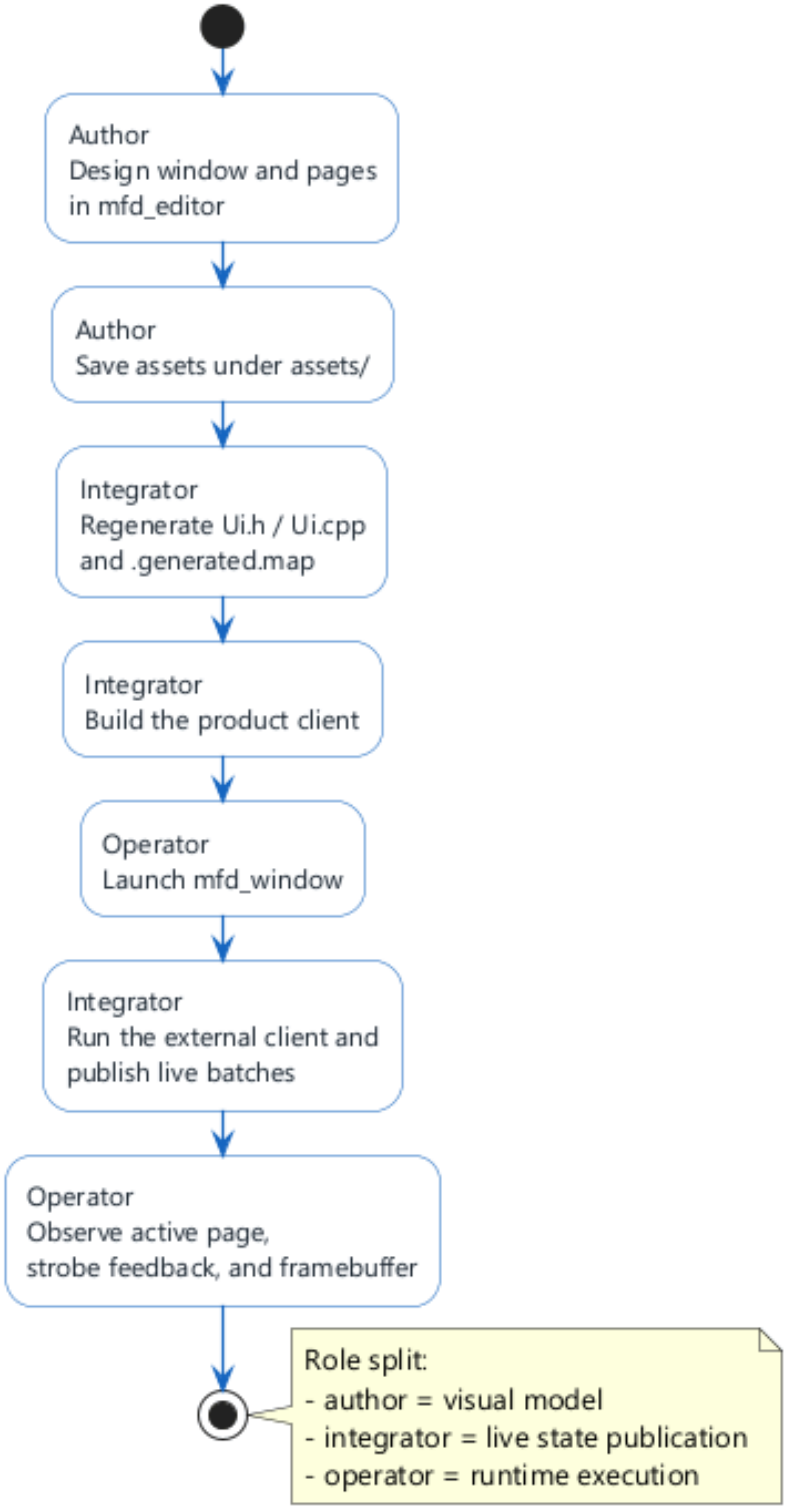
1.1 The main components

Component	Role
mfd_editor	visual tool for authoring windows, pages, and reticles
assets/	versioned source of truth
mfd_client_api/generator	generates typed C++ wrappers and the companion .generated.map file
mfd_runtime_api	dedicated public package for offscreen runtime embedding without exposing mfd_api
mfd_window	runtime host that opens a JSON window, renders the active page, and accepts live commands
client_mockup	reference client that shows how to use the public API
mfd_window_plugin_api	stable C ABI for framebuffer capture DLL plugins

1.2 The minimum mental model



Pipeline authoring to runtime



End-to-end workflow for an author, integrator, and runtime operator

The normal flow of an MFDStudio project is as follows:

1. 1. we author a window and its pages in `assets/` with `mfd_editor`
2. 2. we generate a C++ client API specific to this window
3. 3. we compile a business client which uses this generated API
4. 4. we launch `mfd_window` on the JSON window
5. 5. the client sends UDP commands to the runtime
6. 6. the runtime can send UDP feedback to the client

7. 7. optional, a DLL plugin receives one plugin-selected `RGBA32` or `BGRA32` framebuffer at each frame

For an embedded host application, steps 4 to 7 can also be replaced by one direct `mfd_runtime_api` integration which loads the same authored window, keeps the same UDP in/out contract, and renders into one caller-owned offscreen surface.

1.3 Who does MFDStudio serve?

MFDStudio is suitable when you need:

- cleanly separate graphic authoring and real-time logic
- keep a readable and versionable author model
- drive pages by an external client without embedding the rendering engine in the client
- generate C++ type wrappers from an author window
- embed the runtime offscreen inside a dedicated host application without exposing low-level render dependencies to that application
- plug a framebuffer capture pipeline into a stable DLL ABI

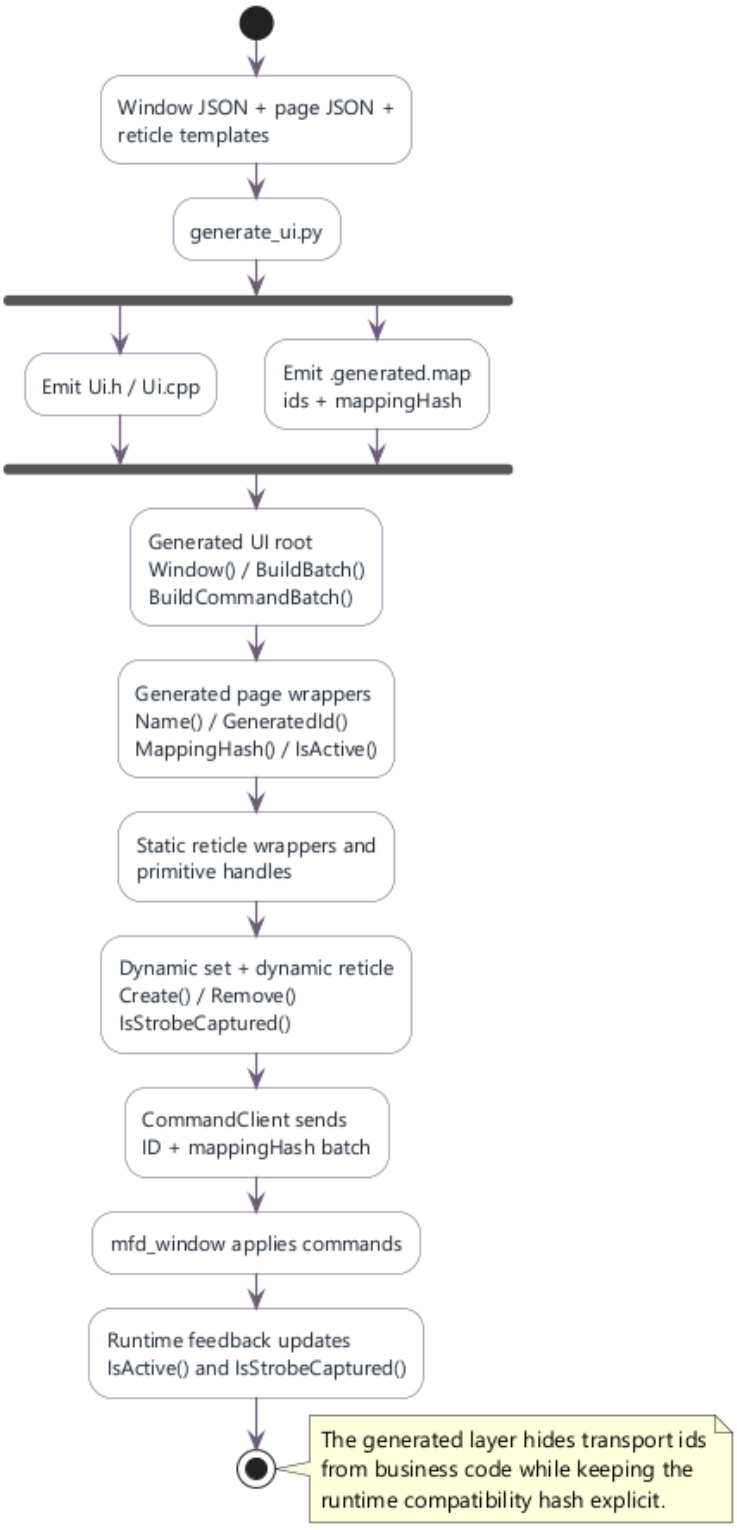
1.4 What MFDStudio is not

MFDStudio is not:

- a general Qt-type UI framework
- a 3D scene authoring system
- a generic network protocol independent of assets
- a plugin system open to any C++ ABI

⚠ WARNING	The runtime remains the owner of the scene state and the rendering. The client remains the owner of the business data and the transmission rate. It is this separation that gives stability to architecture.

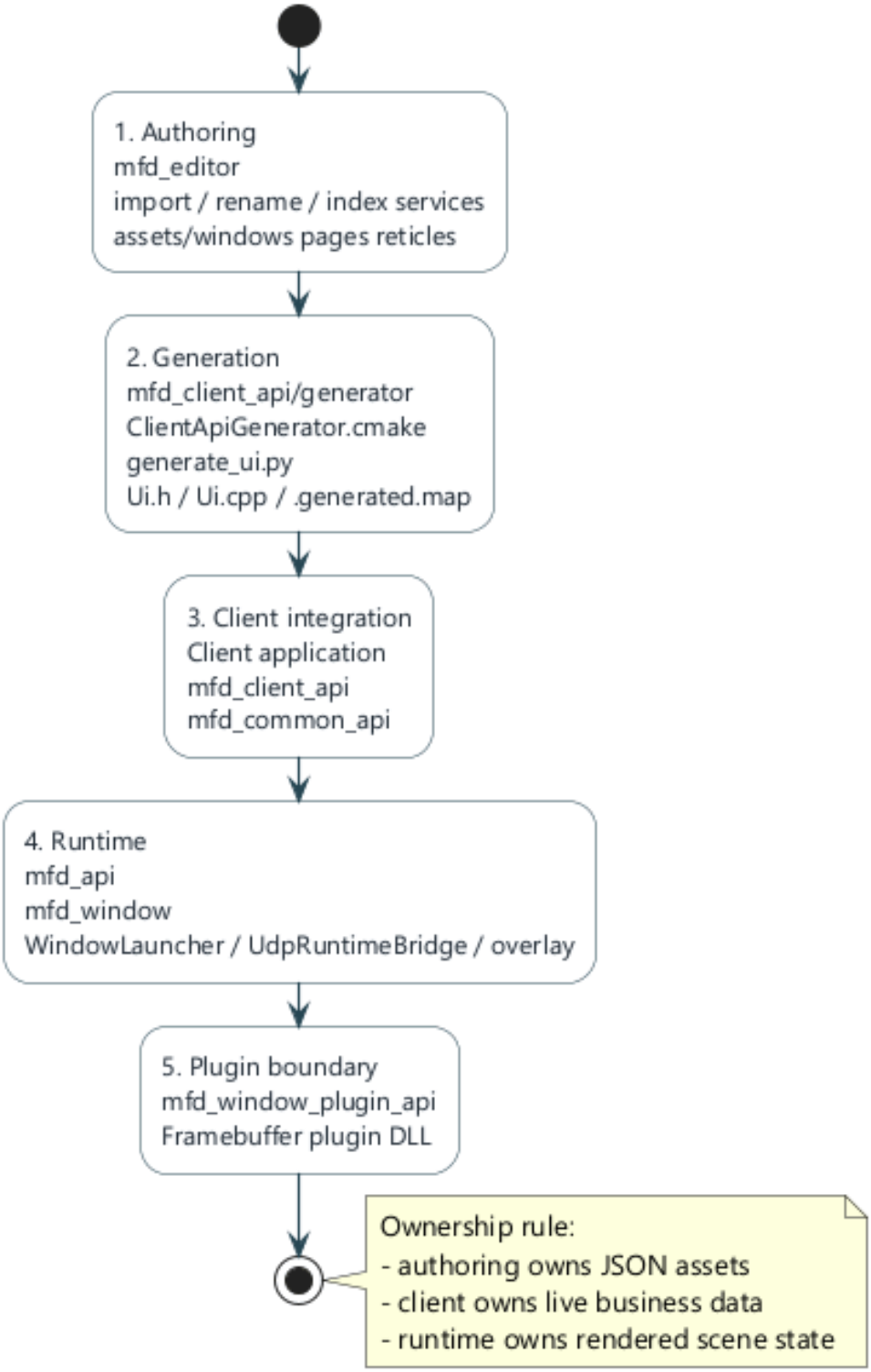
2. Detailed Architecture



API and transport model

The repository architecture is modular. Each module keeps a deliberately narrow responsibility.

2.1 Depot modules



Modular layers of the repository

Module	Primary Responsibility
mfd_common_api	common types, low-level authoring model, transport and commands

mfd_api	loading JSON, runtime, scene registry, and low-level API
mfd_client_api	client overlay: handles, batching, publication, feedback
mfd_client_api/generator	derives a C++ UI and a transport map from a JSON window
mfd_runtime_api	dedicated offscreen runtime package exporting RuntimeSession and OffscreenSurface
mfd_editor	visual authoring and protected workflows on assets
mfd_window	generic runtime launcher and debug overlay
mfd_window_plugin_api	Stable C contract for framebuffer capture plugins
examples/	reference clients and plugins

2.2 Author / runtime / client separation

The architecture is based on three layers which must not be mixed:

Layer	Possessed	Must not own
Authoring	visual structure and JSON assets	live business logic
Runtime	rendering and state authoritarian scene	knowledge of the customer business domain
Customer	live data, cadence, control intentions	the window rendering engine

2.3 The role of the file .generated.map

The generator emits three coherent artifacts:

- a generated C++ header
- a generated C++ source
- a `<window>.generated.map` file

The `.generated.map` contains stable transport IDs for:

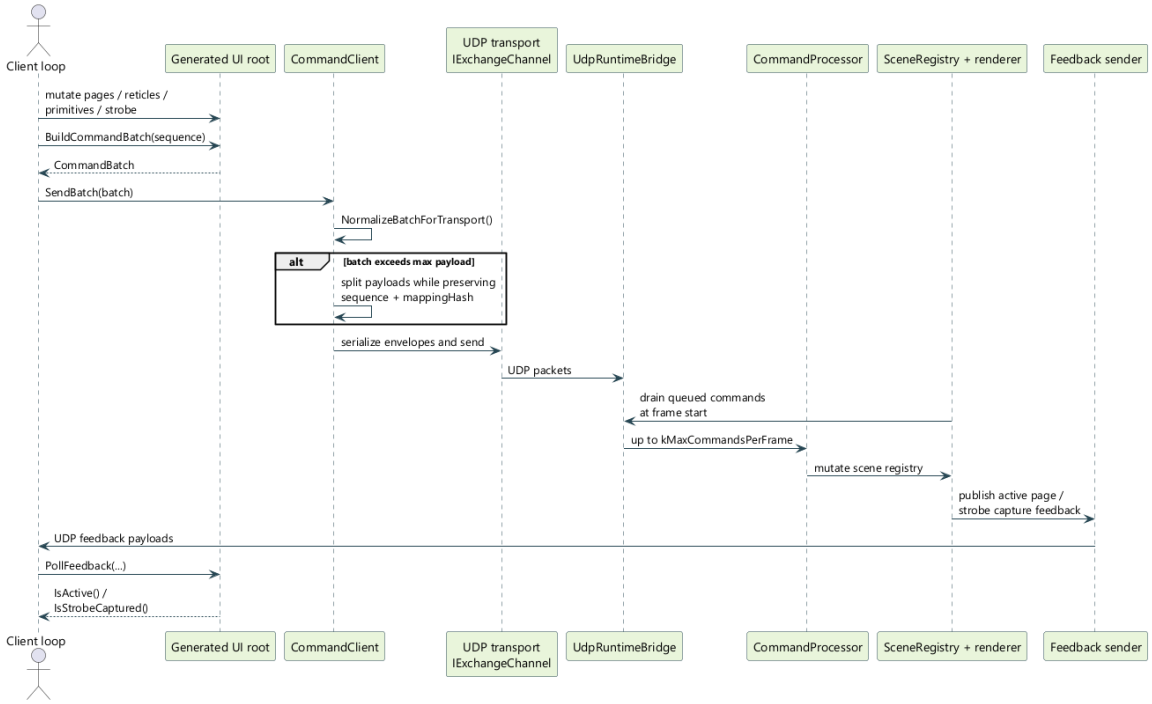
- the pages
- static reticles
- authored strobe reticles
- the primitives exposed

- dynamic templates
- the blink types
- the page-local strobe entries

It also contains a `mappingHash` used to prove that the client and the window are talking about the same author model.

Treat these three outputs as one integration contract. The generated C++ bakes transport ids and `mappingHash` into typed wrappers, while the `.generated.map` must be loaded by both tooling and `mfd_window` when generated id-based batches are used.

2.4 Runtime loop and feedback



Command sequence and feedback

In the recommended mode:

- a UDP worker receives command packets
- the render thread drains commands at the start of the frame
- the runtime applies the mutations via the `CommandProcessor` and the scene
- the runtime can send authoritative feedback to the client

This feedback is used in particular to:

- the active page rendered
- the selected strobe variant and its resolved state
- capturing a dynamic reticle by the active page strobe

2.5 Author model

The minimal author model is:

- a `window`
- a list of `pages`

- of the `staticReticles` instances on each page
- a folder of reusable reticle templates
- of the `dynamicReticleBindings` on the pages
- one optional `strobes` catalog per page plus one optional `activeStrobe`
- of the `blinkTypes` defined at page level

The logical author and client coordinates are normalized in `[-1, 1]`.

2.6 Practical consequences for integration

For a clean integration:

- authoring the visual structure into JSON
- only expose the necessary primitives to the client
- use the generated API rather than text names everywhere
- use `mfd_runtime_api` rather than exposing `mfd_api` directly when one external application must embed the runtime offscreen
- preserve `CommandClient` as final sending layer
- reserve framebuffer plugins for image output, not for business control

3. Prerequisites and tools to build

3.1 Recommended Toolchain

The target deposit in practice:

- Visual Studio 2022
- CMake 3.25 or higher
- C++17
- Python 3 for the generator

3.2 Build local recommends

The preferred local flow is Debug Win32:

CODE POWERSHELL

```
cmake --preset vs2022-win32
cmake --build --preset debug-win32
ctest --preset test-debug-win32
```

3.3 Useful binaries as needed

To author and test the main flow:

CODE POWERSHELL

```
cmake --preset vs2022-win32
cmake --build --preset debug-win32 --target mfd_window mfd_framebuffer_stdout_plugin
client_mockup mfd_editor
```

For one specific generated client:

CODE POWERSHELL

```
cmake --build --preset debug-win32 --target client_mockup_minimal
```

For the dedicated offscreen embedding path:

CODE POWERSHELL

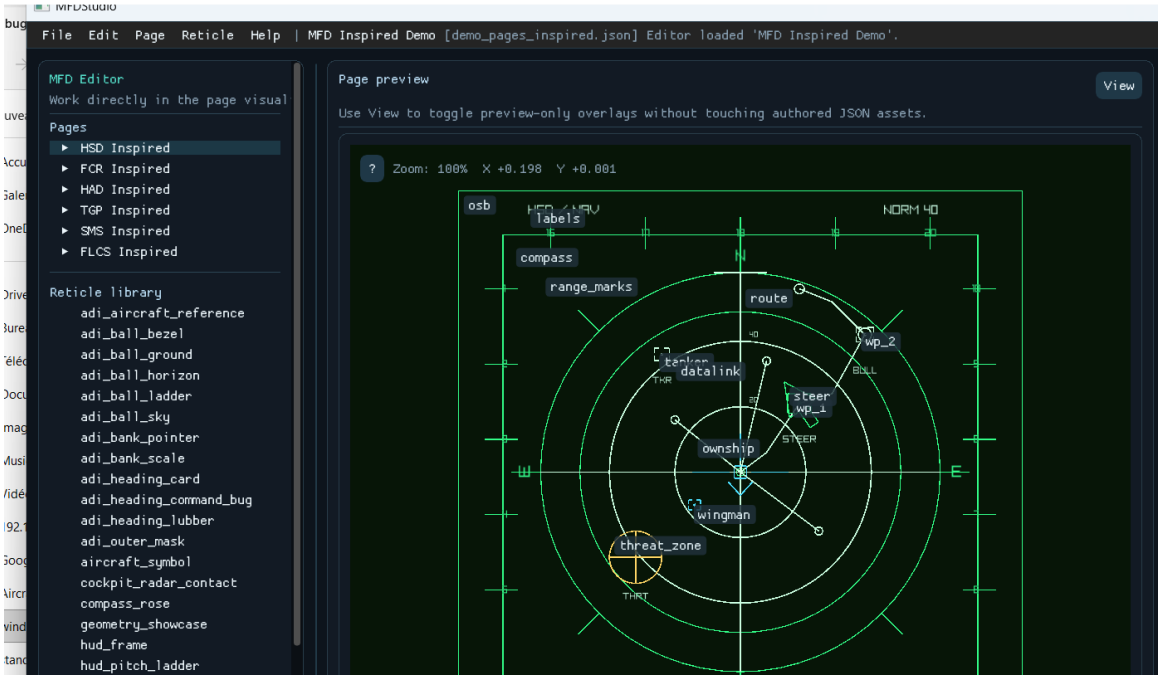
```
cmake --build --preset debug-win32 --target offscreen_viewer
```

3.4 Tree structure to know

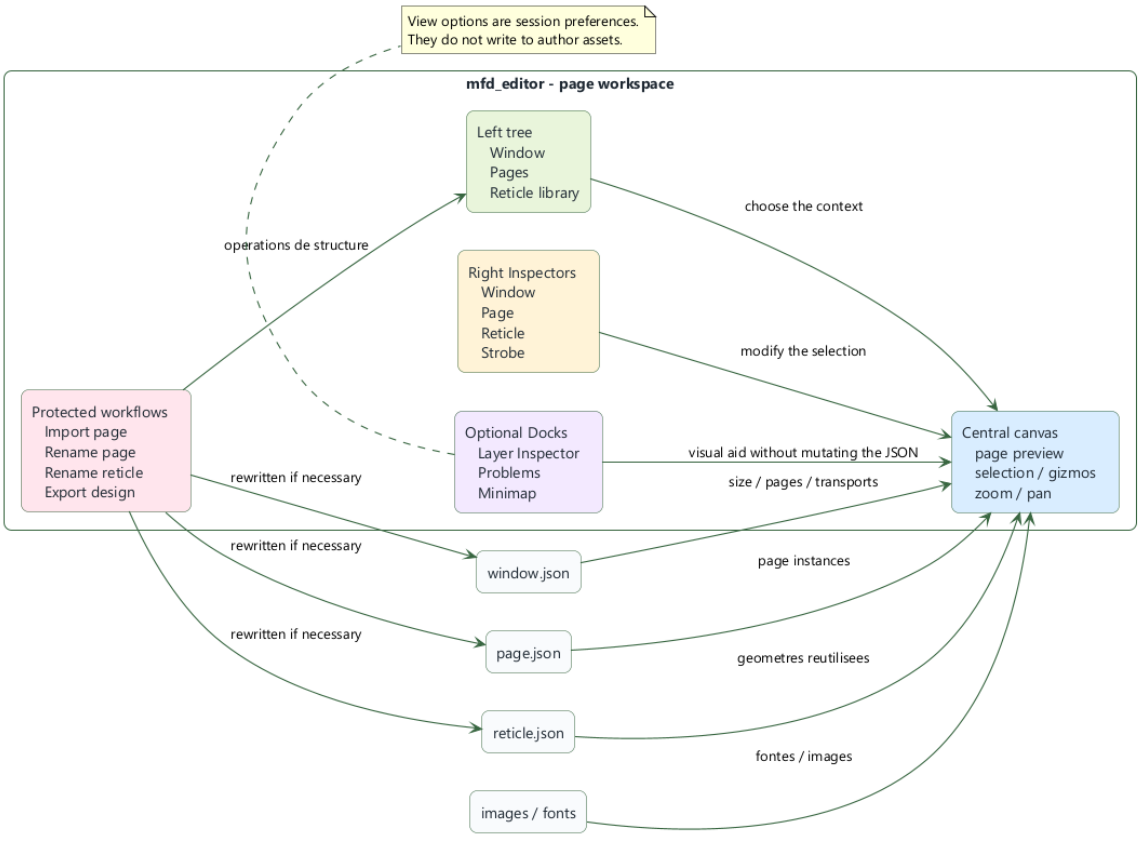
Case	Expected content
assets/windows	JSON root windows
assets/pages	JSON pages
assets/reticles	reticle templates
assets/fonts	possible fonts
examples	reference clients and plugins

Scripts	launch batches
_Exec	runtime staging regenerated by the build
✦ TIP	Author in `assets/` and not in `_Exec`. `_Exec` is a runtime staging area, not the long-term source of truth.

4. Use mfd_editor to design a window and its pages



`mfd_editor` screenshot



Editor workspace

This section describes how to use `mfd_editor` in detail to author a window, its pages, its reticles, and the associated asset-maintenance workflows.

4.1 Launch the editor

Once `mfd_editor` is built:

- launch `mfd_editor.exe`
- or launch the staged executable under `_Exec`
- optionally pass `--asset-directory C:\Path\To\assets` to use another

authored asset root for default window, page, font, and reticle paths

At startup, the editor does not automatically reload a copy from `_Exec`. It opens an empty document until the user:

- load an existing window
- or create a new window
- or reopen a recently authored window

The empty workspace acts as a resume hub: besides the `Open window`, `New window` and `Tutorial` actions, a `Recent` action appears as soon as at least one previously opened window still exists on disk. The same list is also available under `File > Open recent window`. The recent list is editor-local and stored next to the assets in `assets/.editor_recent_windows.json`.

4.2 Anatomy of the interface

The editor is organized around four areas:

Area	Use
left shaft	window, pages, reticle library
central canvas	page preview, selection, gizmos, zoom/pan
straight inspectors	contextual window editing, page, reticle, strobe
optional docks	Layer Inspector, Problems, Minimap

The preview remains the main surface. The Dockes panels reserve a real space and must not hide the canvas.

4.3 Create a new window

Menu:

- `File > New window from scratch`

The popup allows you to enter:

Field	Sense
Window file	path of the JSON window to create
Window title	title visible from window

Size (px)	window width/height
Position (px)	initial screen position
Font file (optional)	font used for text rendering
Reticle library folder	reusable templates folder

These fields correspond directly to the root of the JSON window.

The creation popup uses native Windows pickers: save-file pickers for new window and page JSON files, an open-file picker for the optional font, and a folder picker for the reticle library. The popup stays open after each picker returns so the draft can be completed in one pass.

Once the window is open, the right inspector mirrors those pickers for live editing through `Browse font file...`, `Browse reticle folder...`, and (for an image primitive) `Browse image file...`, so a path no longer has to be typed by hand.

4.4 Configure UDP transports

The editor exposes two configuration groups.

Incoming commands to window

Option	Effect
Enable command UDP	activates the command entry point
Command address	IPv4 listening or link address
Command port	UDP port
Command max packet	max size of a datagram

Feedback coming out of runtime to clients

Option	Effect
Enable feedback UDP	activates the return flow
Feedback address	target IPv4 address
Feedback port	UDP feedback port
Feedback max packet	max datagram size

For a single position, use `127.0.0.1`.

When the feedback UDP flow is enabled, the window also emits a window lifecycle heartbeat and a final `Closing` payload at shutdown. Clients use these to detect when the window is gone and reset their connections (see section 6.20).

4.5 Create a first page

During window creation, you can activate `Create one initial page` then enter:

Field	Sense
First page name	public page name
First page title	human title
First page file	JSON page path
First page background	background color

This first page becomes the `defaultPage`.

4.6 Save what the editor writes

The document first exists in memory. Nothing persists until the user does:

- `File > Save`

The writing may concern:

- the JSON window
- the JSON of the initial page
- reticle templates if the workflow has created them

4.7 Add, choose and organize pages

To add a page:

- `Page > New page`

To choose the default page:

- select the page in the tree
- enable `Default page for this window` in the page inspector

4.8 Design of a page: title chrome, layers, static reticles, dynamic bindings, strobes

Page title chrome

Each page owns one generated title chrome driven by the page `title` text plus an optional `titleDisplay` block.

That authored chrome can:

- stay visible or become invisible
- move inside the page space
- scale and rotate
- change color
- switch decoration between `none`, `underline`, and `frame`

- change the decoration line style and line width

In `mfd_editor`, this chrome is selected from the page inspector through `Select title chrome`, then edited from its dedicated inspector. It is not a separate static reticle asset, but it behaves like one editor-side chrome element for selection and transform gestures.

Page layers

Each page declares `layers` orders. They are the runtime source of truth for the drawing order.

Static reticles

A page can contain `staticReticles` :

- either library template instances
- either reticles inline

Each instance aims at `layerId`.

Dynamic reticle bindings

Section `Dynamic reticles` in the page inspector:

8. 1. choose one `Reticle template`
9. 2. choose one `Layer`
10. 3. to click `Add`

Each entry creates a runtime binding, not a static instance drawn by default. The live client will then create the real runtime instances.

Page strobes

A page can expose one or more named `strobes` plus one authored `activeStrobe` selection.

The editor allows you to:

- add or remove page strobe entries
- choose the authored active strobe for page startup
- assign one template per strobe entry
- edit each entry position and capture/magnetization configuration
- expose individual primitives inside the strobe template from the primitive inspector
- disable parent reticle rotation and scale inheritance per primitive when one label or marker must stay upright

The active entry behaves as the live page strobe until the client selects another authored strobe variant at runtime.

4.9 Important page fields

The JSON page supports in particular:

Canonical field	Use
name	public page name
title	human title
titleDisplay	authored visibility and styling of the generated title

	chrome
backgroundColor	bottom
layers	ordered runtime layers
dynamicReticleBindings	dynamic templates allowed on the page
blinkTypes	page blink types
defaultBlink	blink by default
view.center	center of view
view.zoom	zoom
staticReticles	static reticles
activeStrobe	authored startup strobe selection
strobes	optional named strobe catalog

4.10 Supported primitives for constructing reticles

The families of primitives supported by the author model are:

Primitive	Typical usage	Specific fields
text	labels, values	text, fontSize, letterSpacing, align
time	clock	format, utc, fontSize, letterSpacing, align
line	features	start, end
circle	markers	radius
ring	crowns	innerRadius, outerRadius, segments
rectangle	panels	width, height, size
ellipse	oval markers	width, height, radiusX, radiusY
square	square markers	size, width, height
diamond	waypoints	size, width, height

triangle	pointers	points
polyline	roads, frames	points, closed
bezier	curves	controlPoints, segments
arc	sectors, arcs	radius, startAngleDegrees, endAngleDegrees, segments
image	bitmaps	file, width, height, size

Common primitive fields cover:

- `position`
- `rotationDegrees`
- `scale`
- `visible`
- `exposed`
- `reticleRotationSensitive`
- `reticleScaleSensitive`
- `stroke`
- `thickness`
- `lineStyle`
- `filled` for fill-capable primitives
- `fill` for fill-capable primitives

Fill-capable means circles, rings, rectangles, ellipses, squares, diamonds, triangles, arcs, and polylines when `closed` is `true`. The editor persists `filled` explicitly as `true` or `false` for those cases and omits it for non-fillable primitives.

`reticleRotationSensitive` and `reticleScaleSensitive` default to `true`. Disable them when one primitive must ignore parent page-reticle or page-strobe rotation and scale while still remaining patchable directly through the generated client API.

`align` is supported on `text` and `time` with three values only: `left`, `center`, and `right`. `center` remains the default and preserves the historical behavior. In the editor, changing only the alignment does not move the currently authored text immediately; it changes which side expands later when the rendered string becomes wider.

4.11 Selection and direct editing on the canvas

The page preview supports:

- `Ctrl+click` to add or remove a reticle from the selection
- `Esc` to empty the selection
- dragging a selection to move a group
- `Ctrl+C`, `Ctrl+X`, `Ctrl+V`, `Del`
- toolbar `R` button to recenter the preview camera on the authored page view

The reticle studio supports:

- clicking one primitive to focus it in the inspector

- dragging gizmos to edit the selected primitive geometry
- `Ctrl+C` / `Ctrl+V` to duplicate the focused primitive inside the current reticle template
- `Ctrl+C` / `Ctrl+V` with no primitive focused to copy and paste the selected library reticle template
- toolbar `R` button to recenter the studio camera

In case of superimposed reticles:

11. 1. right click in the area
12. 2. choice of the desired reticle in the context menu
13. 3. copy/cut/delete operations on the current selection

4.12 The View menu

The `View` menu lives in the top menu bar and stays available at all times. Its content adapts to the active workspace; even before a window is open it still exposes the `Panels` sidebar and inspector toggles.

In the page-preview and fullscreen workspaces, the available toggles are:

Option View	Effect
Panels > Sidebar	shows or hides the left page and reticle selector (persisted)
Panels > Inspector	shows or hides the right inspector (persisted)
Layer Inspector	layer dock and focus by layer
Minimap	mini navigation view
Problems	dockes diagnostics
Highlight reticle usages	highlights the pages that reference the selected template
Reticle names	displays names
Gizmos	shows visual manipulations
Fullscreen preview	enters or leaves fullscreen preview (F11)

While editing a library reticle, the same menu instead exposes `Show page context`, `Show primitive names`, `Show gizmos`, plus the shared `Grid`, `Snap to grid`, and `Grid step` controls.

These options are session preferences. They do not rewrite the author JSON.

Fullscreen preview

The `Fullscreen preview` entry in the `View` menu, the `F11` key, or `Esc` allows you to:

- hide sidebar, inspectors and docks
- keep the canvas interactive
- restore the previous state then

It is a pure editor layout mode, available in every workspace and from the moment the editor opens, even before a window is loaded.

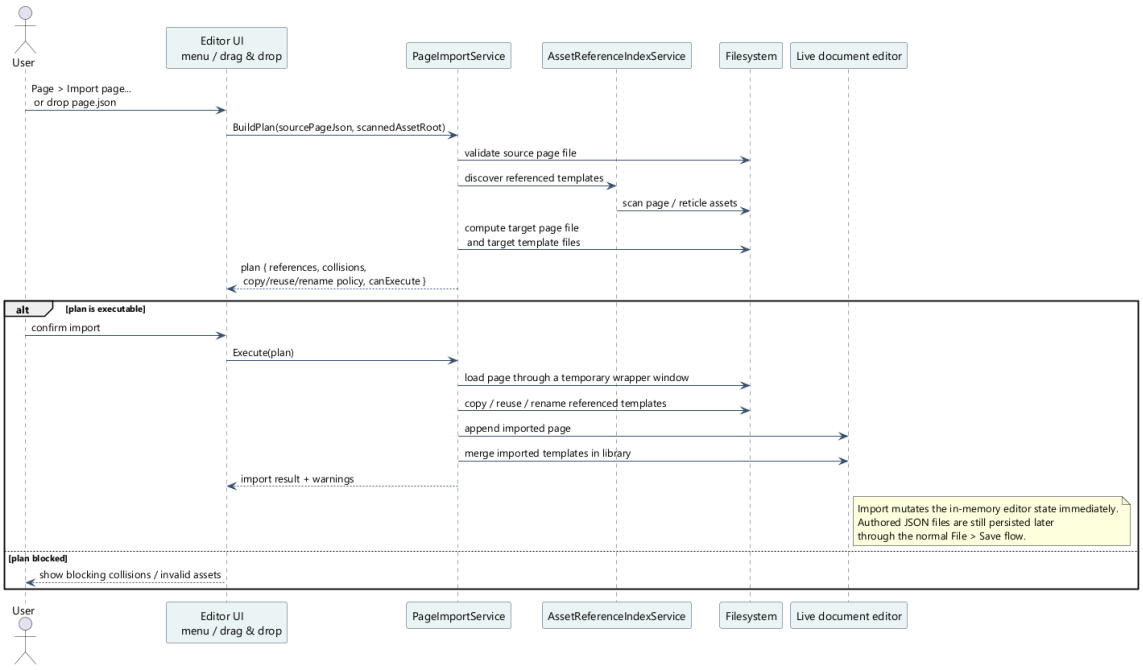
4.13 Layer Focus Mode

THE `Layer Inspector` allow :

- duty `Full View`
- to list the runtime layers
- to display a thumbnail and a counter per layer
- to focus a layer to leave only its reticles editable

The other visible layers remain rendered, but dimmed.

4.14 Import an existing page



Page import sequence in the editor

Entries:

- `Page > Import page...`
- or drag and drop a page JSON

The workflow calculates a plan before execution:

- source page
- target in current tree
- template references
- collisions
- copy/reuse/rename strategy

The current rules are:

- target file missing: copy
- identical file already present: reuse
- different file already present: creation of a renamed copy

4.15 Delete or detach a page

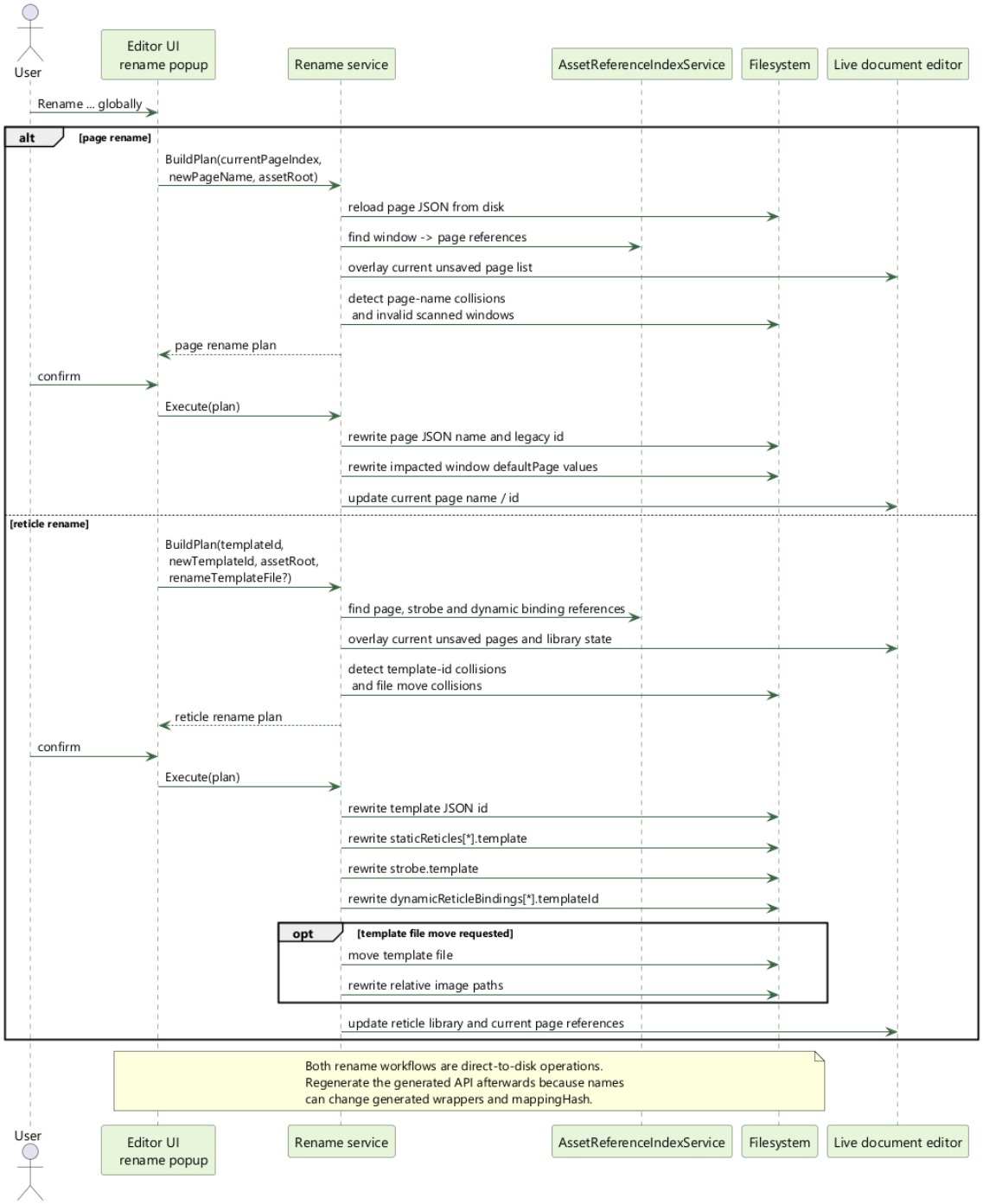
Two distinct actions exist:

Action	Effect
Remove page from window	removes the page from the current window but keeps the file
Delete page asset…	removes the page and marks its JSON for deletion

Security guards:

- the window must keep at least one page
- we do not delete the default page without choosing another one
- deletion out `assets/` blocked without explicit override

4.16 Properly rename a shared page



Global page/reticle rename sequence

Entries:

- right click page `Rename page globally...`
- inspector button
- menu `Page > Rename current page globally...`

The rename service:

- rescan references `window -> page`
- rewrites the page JSON
- rewrite the `defaultPage` impacts
- blocks collisions before any mutation

- warns that the generated API and the `mappingHash` can change

4.17 Properly renaming a shared reticle template

Entries:

- right click template `Rename reticle globally...`
- inspector button
- menu `Reticle > Rename selected library reticle globally...`

The workflow:

- rewrite the `id` of the template
- rewritten `staticReticles[*].template`
- rewritten `strobes[*].template`
- rewritten `dynamicReticleBindings[*].templateId`
- can also rename the template file
- updates relative image paths if file moves

4.18 Highlight use of a template

Normal flow:

14. 1. select a reticle template in the tree
15. 2. enable `View > Highlight reticle usages`
16. 3. the editor highlights:

- the pages that reference this template
- the corresponding instances on the active page

4.19 Export of design documentation

Menu:

- `File > Export > Export design...`

The exported package may contain:

Exit	Content
README.md	package entry exports
window_icd.md	window view
pages/*.md	detailed views per page
images/*_design_exploded.png	exploded views with reticle labels if rendering was successful
images/*_design_page.png	clean page views without reticle labels if rendering was successful

data/design_manifest.json	design manifesto exports

Options available in the popup:

- Markdown ICD snippets
- exploded views
- contact details
- strobe
- blink
- primitive ids
- hash mapping

4.20 Recommended authoring checklist

17. 1. create the window and its transports
18. 2. define the pages and `defaultPage`
19. 3. declare runtime layers
20. 4. compose static reticles and only expose useful primitives to the client
21. 5. declare the `dynamicReticleBindings` necessary
22. 6. define the `strobes` catalog and `activeStrobe` if the page needs cursor behavior
23. 7. check `Problems`, `Layer Inspector`, `Minimap`
24. 8. save as `assets/`
25. 9. regenerate the API if the exposed runtime surface has changed

5. Generate the client API from the generator

For a C++ integrator, the generator is not a convenience wrapper. It is the compatibility boundary between authored JSON assets and the code that will publish live commands at runtime.

If generation is treated as a real build step, you get a typed, repeatable, window-specific integration contract. If generation is skipped, or if the generated files are stale, you lose the proof that the client and the runtime still agree on the same authored surface.

■ 5.1 Entry, exit, contract

The generator consumes one authored window entry point:

- one root window JSON
- every referenced page JSON
- every reticle template reachable from the window

It emits three integration artifacts:

- one generated C++ header
- one generated C++ source
- one `<window>.generated.map` sidecar

These outputs are one contract, not three independent files. The generated C++ layer bakes generated transport ids and the `mappingHash` into the typed wrappers. The `.generated.map` exposes the same canonical transport surface to the client, the runtime, and any raw-name tooling.

■ 5.2 Official CMake macro

The supported integration path is the `client_api_generate_ui(...)` CMake macro.

```cmake
client_api_generate_ui(
    WINDOW_JSON "assets/windows/demo_pages_cockpit.json"
    OUTPUT_HEADER "${CMAKE_CURRENT_SOURCE_DIR}/generated/MockupUi.h"
    OUTPUT_SOURCE "${CMAKE_CURRENT_SOURCE_DIR}/generated/MockupUi.cpp"
    OUTPUT_MAP "${MFD_ROOT_DIR}/assets/windows/demo_pages_cockpit.generated.map"
    NAMESPACE "mockup_ui"
    UI_CLASS_NAME "CockpitMockupUi"
    HEADER_INCLUDE "MockupUi.h")
```

Do not replace this macro with an ad hoc custom command unless you also reproduce its input-tracking behavior. The difficult part is not launching Python; the difficult part is ensuring CMake knows when authoring assets changed.

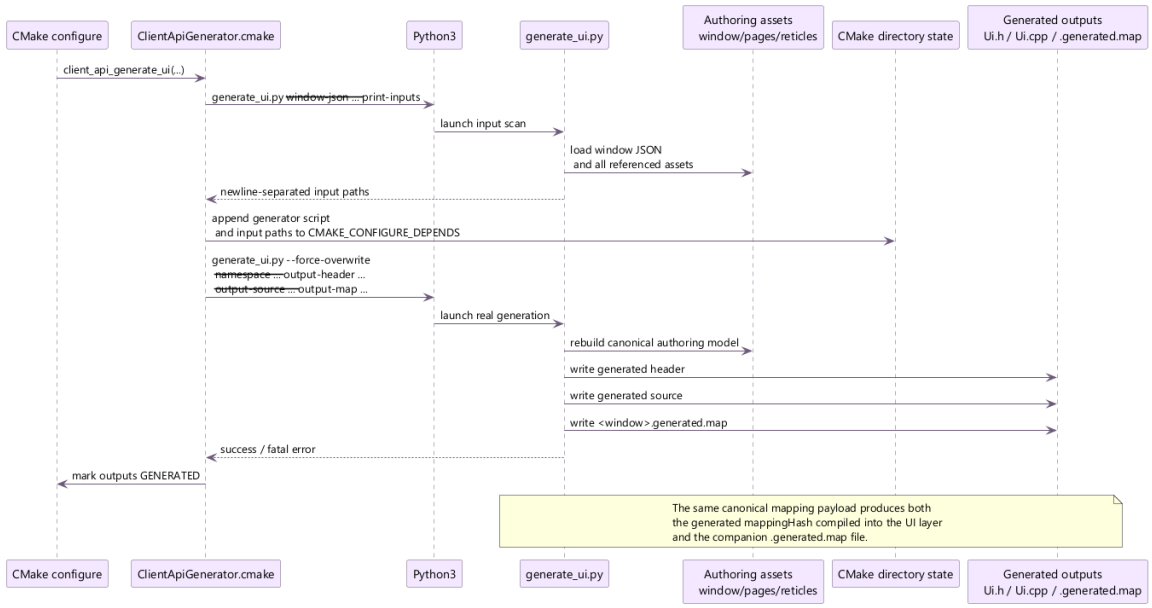
■ 5.3 client_api_generate_ui arguments

Argument	Mandatory	Integration meaning
WINDOW_JSON	yes	authored root window to parse
OUTPUT_HEADER	yes	generated typed API header

OUTPUT_SOURCE	yes	generated typed API source
OUTPUT_MAP	yes	generated transport sidecar required by the generated client/runtime contract
NAMESPACE	no	namespace exposed to the client target
UI_CLASS_NAME	no	explicit UI root class name
PAGE_CLASS_SUFFIX	no	suffix appended to generated page wrapper names
UI_CLASS_SUFFIX	no	suffix appended to the UI root if UI_CLASS_NAME is omitted
HEADER_INCLUDE	no	include path written into the generated .cpp
CONFIGURE_DEPENDS	no	extra paths that must also trigger CMake reconfigure

The important point is that `WINDOW_JSON` is only the entry point. The real dependency set is broader, which is why the input-scan mode exists.

■ 5.4 What the CMake macro really does



Client API generation sequence

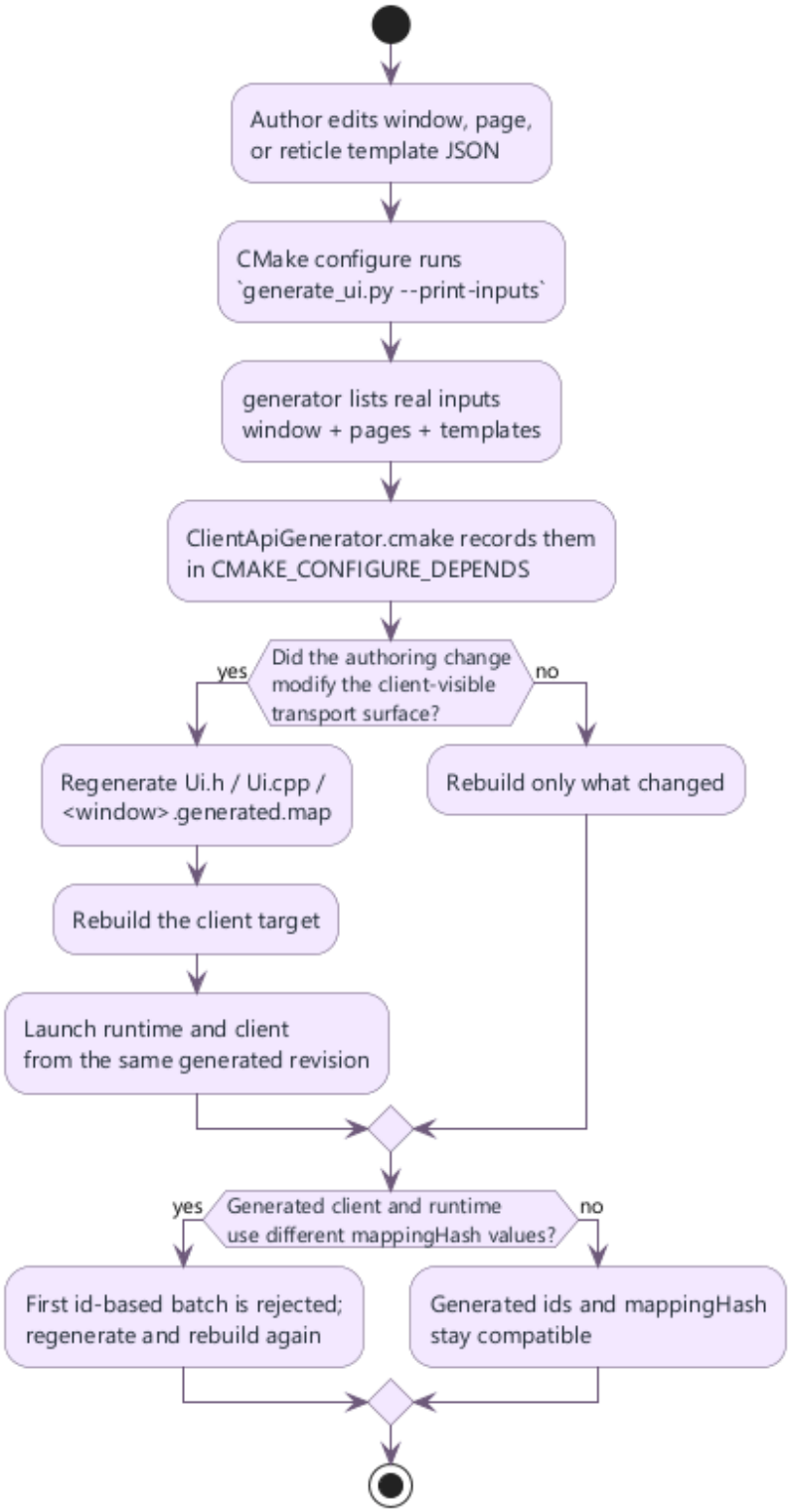
The macro performs two distinct passes:

26. 1. a configure-time scan through `generate_ui.py --print-inputs`
27. 2. a real generation pass through `generate_ui.py --force-overwrite ...`

The input scan discovers the actual dependency graph behind the window:

- the root window JSON
- every referenced page JSON
- every template file in the active reticle library

Those paths are appended to `CMAKE_CONFIGURE_DEPENDS`, which means CMake knows that an authoring change can invalidate the generated client contract even if no C++ file changed.



Developer workflow from authoring change to regenerated client API

5.5 Why --print-inputs matters for incremental builds

The generator supports a discovery mode:

CODE POWERSHELL

```
py -3 mfd_client_api/generator/scripts/generate_ui.py `
  --window-json assets/windows/demo_pages_cockpit.json `
  --output-header generated/MockupUi.h `
  --output-source generated/MockupUi.cpp `
  --print-inputs
```

The output is a newline-separated list of absolute input paths. `ClientApiGenerator.cmake` captures that stdout and registers each path in `CMAKE_CONFIGURE_DEPENDS`.

Why this matters in practice:

- if a page file changes, the generated API is invalidated
- if a reticle template changes, the generated API is invalidated
- if a new dynamic template binding is added to a page, the generated API is invalidated

Without `--print-inputs`, CMake would only see the top-level generator invocation. It would not know that editing a page or template changed the transport surface behind the generated code.

⬡ API	`--print-inputs` is part of the build contract, not a debugging option. Removing it from the CMake path makes stale generated code much more likely.

5.6 What breaks if you do not regenerate

The most visible failure mode is a `mappingHash` mismatch after an authored change that altered the transport surface:

- page rename
- static reticle rename
- exposed primitive rename
- dynamic template binding added or removed
- blink type rename

Typical sequence:

28. 1. an author changes the authored model
29. 2. `mfd_window` loads the new window JSON and the new `.generated.map`
30. 3. the client executable is still compiled against the old generated `Ui.cpp`
31. 4. the first id-based batch is rejected before the runtime mutates any scene state

Concrete example:

CODE DIFF

```
// Scenario: the client still ships yesterday's generated UI after a page rename.
- batch.mappingHash = "3852bb1a1250284ed4db4ed38d08ea5da4a2044632441bc7afdac7d5cf5885e4";
+ batch.mappingHash = "91a4d9cd12cb6b75c96c0f8f1a3d8d9f6dd8c0f6d2d9bb4a60ee8d8f4261d4ab";

- runtime error: "Generated transport map hash mismatch between the client batch and the
runtime window"
+ after regeneration: batch accepted and generated ids resolve again
```

Two related mismatch cases matter operationally:

Failure point	Exact symptom	What it usually means
client-side normalization	Command batch mappingHash does not match the generated transport map configured on the client	the executable loads a newer .generated.map but still embeds older generated C++
runtime command processor	Generated transport map hash mismatch between the client batch and the runtime window	the client batch was built from a different generated revision than the runtime window

If the transport surface changed, the only correct fix is:

32. 1. regenerate `Ui.h`, `Ui.cpp`, and `.generated.map`
33. 2. rebuild the client target
34. 3. relaunch the runtime with the same authored asset revision

■ 5.7 Rules for exposing primitives

The generated API should model operational control, not authoring noise.

Good exposure candidates:

- labels updated from live avionics data such as heading, Mach, or threat text
- geometry that the client must steer at runtime such as a command bug, a cue line, or a steering symbol
- dynamic template families that represent tracks, threats, waypoints, or cues

Bad exposure candidates:

- decorative lines that never move independently
- duplicated framing labels
- unstable implementation-only primitives that have no meaning to the integrator

Practical rule: if the client code would never mention the primitive in a requirements review, do not expose it.

■ 5.8 What the generated API contains

The generated output builds one typed object tree per authored window:

- one UI root class such as `CockpitMockupUi`
- one page wrapper per page
- one static reticle wrapper per static reticle
- one specialized handle per exposed primitive
- one strobe-reticle wrapper per authored strobe entry
- one generated dynamic-reticle set per page/template binding
- one optional page-scoped `strobe` handle plus generated authored strobe entries when the page defines several variants
- one feedback layer that drives `IsActive()` and `IsStrobeCaptured()`

The generated wrappers intentionally keep the raw label-based support API out of the normal user surface. Application code is expected to stay on generated members such as

`ui.Cockpit().adiHeadingBox.HeadingValue()` or `ui.Radar().DynamicRadarTrack().Create()` instead of calling generic label-driven helpers.

This is why the generated path should be the default integration path. The business code manipulates typed handles and leaves transport ids, runtime dynamic ids, and patch serialization to the generated layer.

■ 5.9 What the .generated.map sidecar contains

The `.generated.map` file is not redundant output. It is the canonical transport description used by both the client and the runtime.

Table	Runtime meaning
pages	page transport ids and default page marker
reticles	static reticle ids keyed by page
primitives	exposed primitive ids keyed by static reticle, strobe reticle, or template
templates	generated ids for dynamic templates
blinkTypes	generated ids for per-page blink types
strobes	generated ids for page-local strobe entries
mappingHash	canonical compatibility fingerprint for the whole transport surface

Keep the sidecar next to the authored window. When the client also needs raw-name helpers, the same map lets `CommandClient` normalize names and detect stale compiled code.

■ 5.10 Example of integration into a CMake target

CODE CMAKE

```
client_api_generate_ui(
    WINDOW_JSON "assets/windows/demo_pages_cockpit.json"
    OUTPUT_HEADER "${CMAKE_CURRENT_SOURCE_DIR}/generated/MockupUi.h"
    OUTPUT_SOURCE "${CMAKE_CURRENT_SOURCE_DIR}/generated/MockupUi.cpp"
    OUTPUT_MAP "${MFD_ROOT_DIR}/assets/windows/demo_pages_cockpit.generated.map"
    NAMESPACE "mockup_ui"
    UI_CLASS_NAME "CockpitMockupUi"
    HEADER_INCLUDE "MockupUi.h")

add_executable(client_mission
    src/main.cpp
    generated/MockupUi.cpp)

target_include_directories(client_mission
    PRIVATE
        ${CMAKE_CURRENT_SOURCE_DIR}/generated)
```

```
target_link_libraries(client_mission
    PRIVATE
        mfd_client_api)
```

The generated `.cpp` belongs in the target sources. Do not treat the generated layer as header-only: the source file carries the baked `mappingHash`, generated ids, feedback glue, and batch builders.

For standalone clients and generated code, `mfd_client_api` is the only MFD library that should appear on the link line. Do not add `mfd_common_api` or `mfd_api` to customer-facing client applications.

The packaged client SDK is curated accordingly: it ships `mfd_client_api`, `ClientSdk.h`, the republished loader/transport/projection headers they require, and the generator helpers, but it does not ship `mfd_api.dll`, `mfd_api.lib`, `SceneRegistry.h`, `CommandProcessor.h`, or other host-runtime headers.

`ClientSdk.h` stays focused on standalone helpers. Generated window headers keep including `mfd/client/GeneratedUiSupport.h` directly so the generated surface does not rely on broad transitive includes.

▪ 5.10.1 Example of packaged consumption from _Deliveries

The repository now carries one reference target, `examples/client_test_package`, whose job is to validate the package boundary itself. It does not link in-tree MFD targets directly; it consumes the staged delivery exactly like an external customer project would:

CODE CMAKE

```
find_package(MFDStudioClientApi REQUIRED CONFIG
    PATHS "${MFD_ROOT_DIR}/_Deliveries/packages_windows/MFDStudioClientApi/build/native"
    NO_DEFAULT_PATH)

client_api_generate_ui(
    WINDOW_JSON "${MFD_ROOT_DIR}/assets/windows/package_test_window.json"
    OUTPUT_HEADER "${CMAKE_CURRENT_SOURCE_DIR}/generated/PackageTestUi.h"
    OUTPUT_SOURCE "${CMAKE_CURRENT_SOURCE_DIR}/generated/PackageTestUi.cpp"
    OUTPUT_MAP "${MFD_ROOT_DIR}/assets/windows/package_test_window.generated.map"
    NAMESPACE "package_test_ui"
    UI_CLASS_NAME "PackageTestUi"
    HEADER_INCLUDE "PackageTestUi.h")

target_link_libraries(client_test_package
    PRIVATE
        MFDStudio::ClientApi)
```

Important consequences:

- the client runtime payload is `mfd_client_api.dll` only
- a packaged client does not need `mfd_api.dll` on its side
- the package config exposes only the build configurations actually delivered

This means a local `Debug`-only delivery can validate `client_test_package` without also generating a `Release` delivery, and inversement.

Typical local validation flow:

CODE POWERSHELL

```
.\Scripts\BuildDeliveries.bat --version 1.8.5 --preset debug-win32-no-tests
cmake --preset vs2022-win32-no-tests
cmake --build --preset debug-win32-no-tests --target client_test_package
```

5.11 When to regenerate

Regenerate whenever the authored transport surface changes.

Must regenerate:

- page names
- static reticle ids
- exposed primitive ids
- template ids used by dynamic bindings
- dynamic page bindings
- page blink types
- the default page when startup behavior depends on it

Usually does not require regeneration:

- purely decorative geometry changes that do not affect exposed ids
- static color or layout changes on content that is not exposed to the client
- runtime logic changes inside the client executable only

Safe rule: if the client-visible contract changed, regenerate before the next compile.

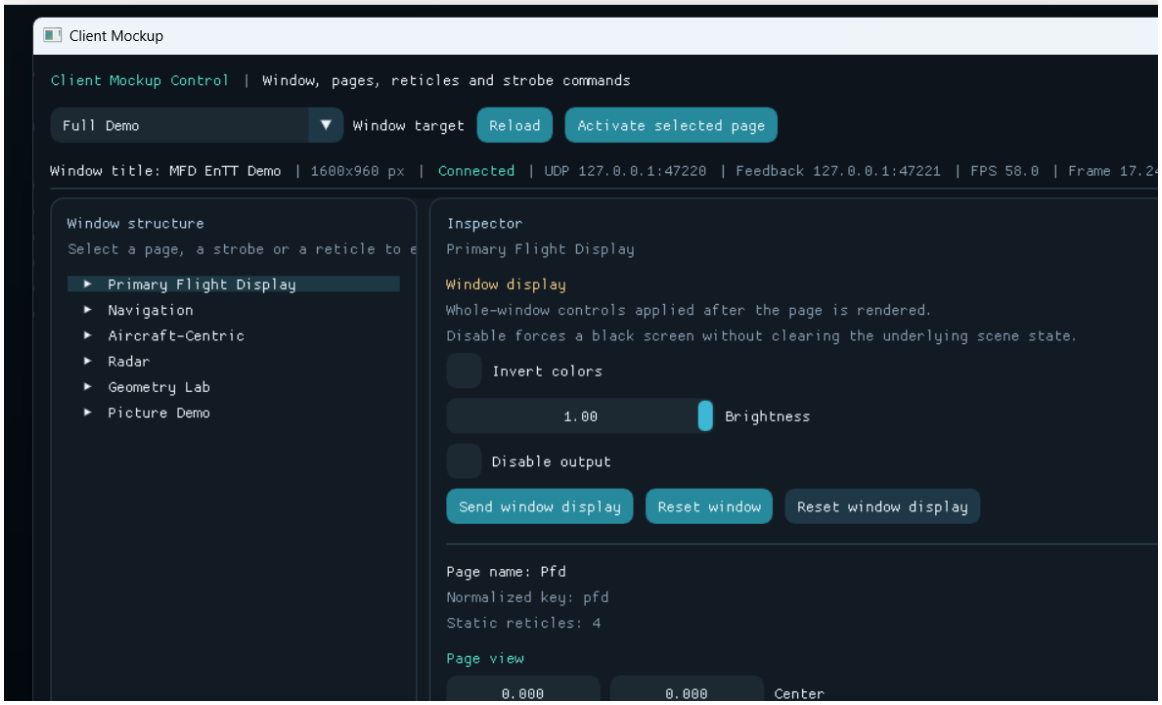
5.12 Troubleshooting the generator

Symptom	Likely cause	Fix
generated header does not expose a new accessor	the primitive was not exposed or the build reused stale outputs	expose the primitive if needed, then rerun generation
runtime rejects the first batch with a hash mismatch	generated C++ and runtime assets are from different revisions	regenerate outputs, rebuild the client, relaunch with matching assets
CMake did not rerun generation after a template edit	the build bypassed client_api_generate_ui(...) or --print-inputs was removed	restore the official macro flow
generator fails with duplicate generated names	two authored ids normalize to the same C++ name	rename the authored ids to distinct stable names
generator fails on unknown dynamic layer	dynamicReticleBindings[*].layerId references a layer not declared in the page	declare the layer or fix the binding
generator refuses to overwrite outputs	the Python entry point was run directly without --force-overwrite	use the CMake macro or pass --force-overwrite intentionally

⚠ WARNING

The generator is not cosmetic. It defines the transport contract between authored assets, runtime, and client code. Any authoring change that affects that contract must be treated like an interface change.

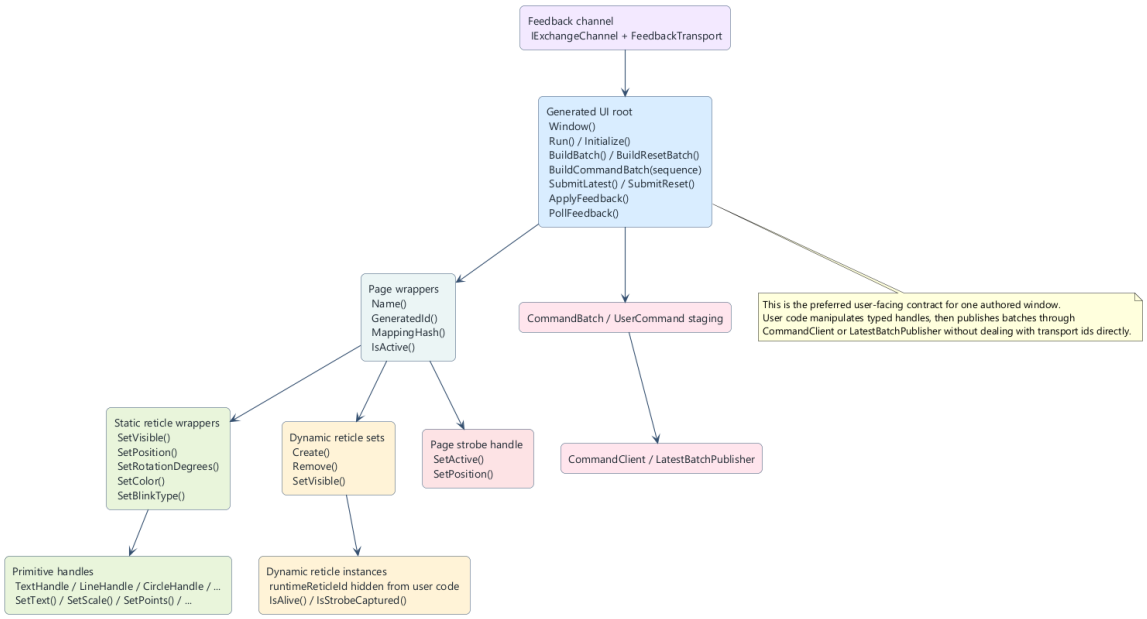
6. Use the generated API to communicate with a window



`client_mockup` runtime client screenshot

This chapter focuses on the recommended client pattern for a real integration: local state mutation first, coherent batch publication second, feedback-driven gating third.

6.1 Two layers to keep separate



Generated API user contract

Keep these two layers conceptually separate:

- the generated typed API, which models one authored window as pages, reticles, primitive handles, dynamic sets, and feedback helpers

- the transport layer `mfd::CommandClient`, which serializes and sends `UserCommand` or `CommandBatch` payloads

The intended use is:

35. 1. mutate generated handles locally
36. 2. let the generated layer collect only dirty state
37. 3. build one `CommandBatch`
38. 4. publish it through `CommandClient` or `LatestBatchPublisher`

⚠ WARNING	Wrong: call `client.Send(...)` or `client.SendBatch(...)` after every `SetText()` and `SetPosition()`. Right: update the full frame locally, then send one coherent batch for that frame.

■ 6.2 Load the authored window, transport, and feedback channel

Start from the real authored window file emitted by the project:

CODE CPP

```cpp
// Scenario: bootstrap a cockpit client that will drive HUD and radar content.
#include "MockupUi.h"
#include "mfd/client/ClientSdk.h"

mfd::JsonLoader loader;
const mfd::LoadedWindowConfiguration loaded =
    loader.LoadWindowConfiguration(std::string(mockup_ui::CockpitMockupUi::WindowFile()));

if (!loaded.window.commandTransports.udp.has_value())
{
    throw std::runtime_error("The window does not expose an enabled UDP command transport");
}
if (!loaded.generatedTransportMap.has_value())
{
    throw std::runtime_error("The window does not expose a generated transport map");
}

std::unique_ptr<mfd::IExchangeChannel> feedbackChannel;
if (loaded.window.feedbackTransports.udp.has_value())
{
    feedbackChannel =
mfd::CreateFeedbackReceiverChannel(*loaded.window.feedbackTransports.udp);
}
```

`mfd/client/ClientSdk.h` is the supported umbrella include for standalone client applications. It keeps example integrations on the `mfd_client_api` SDK surface while still exposing the raw transport helpers when they are needed. The generated window header remains the dedicated entry point for the typed UI surface itself.

`OUTPUT_MAP` is part of the required generated contract. At runtime, `mfd_window` must also load the matching sidecar next to the window JSON. If that matching map is absent, generated id-based batches are rejected because the runtime cannot validate the embedded `mappingHash` and generated transport ids.

6.3 Create CommandClient from the generated transport map

CODE CPP

```
// Scenario: create the transport bridge used by the cockpit control loop.
mfd::CommandClient client(
    *loaded.window.commandTransports.udp,
    loaded.generatedTransportMap);

if (!client.IsReady())
{
    throw std::runtime_error("Unable to initialize the command client: " +
client.LastError());
}
```

`CommandClient` owns:

- transport readiness
- serialization
- UDP payload splitting when one batch exceeds the configured packet size
- normalization of generated ids and named authored ids
- compatibility checks against the loaded transport map

6.4 UI root, startup, and authored default view

The generated UI root centralizes the typed surface for one authored window:

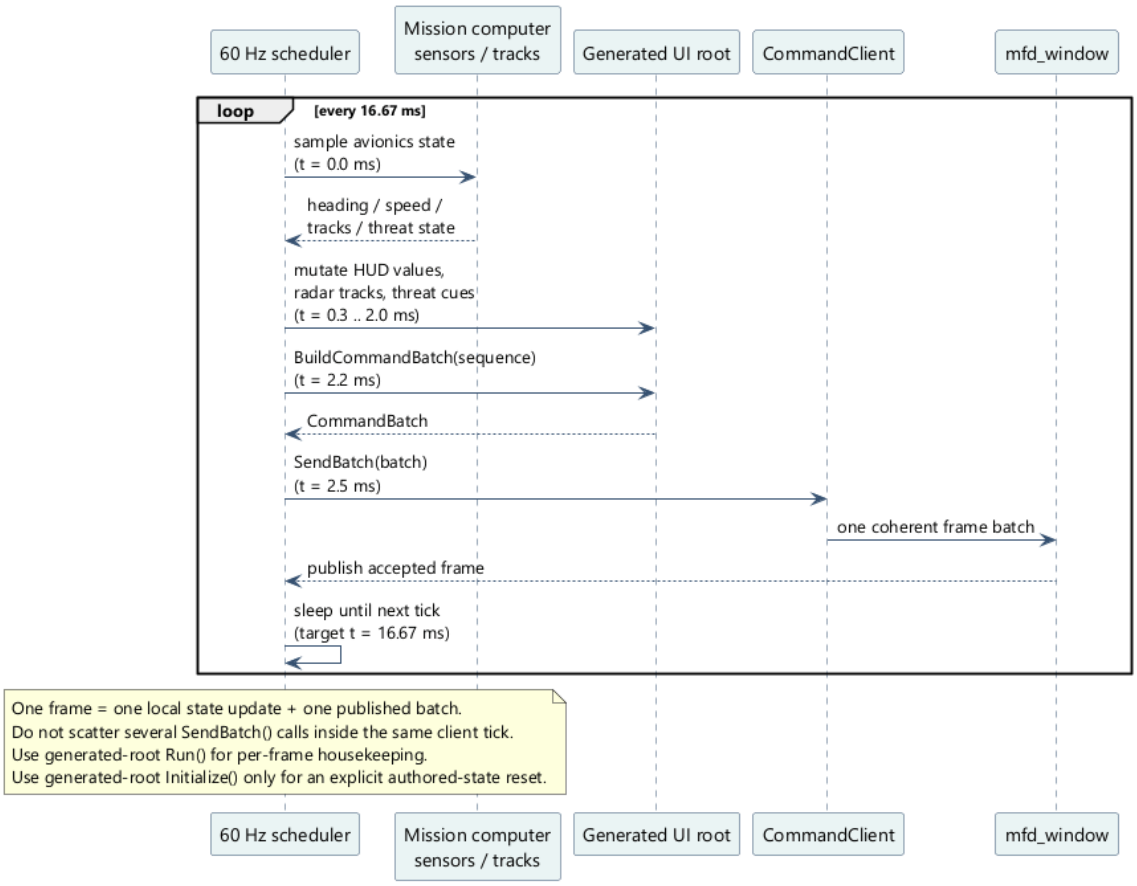
CODE CPP

```
// Scenario: prime the cockpit window before entering the realtime loop.
mockup_ui::CockpitMockupUi ui;
auto& cockpit = ui.Cockpit();

if (!ui.SendStartup(
        client,
        mfd::PageViewState {{0.0f, 0.0f}, 1.0f},
        std::string {"WP-03 READY | awaiting first radar frame"}))
{
    throw std::runtime_error("Unable to send startup batch: " + client.LastError());
}
```

`SendStartup(...)` is useful when your integration needs one predictable initial page, one predictable authored view, and one initial status caption before the continuous loop starts.

⚠ WARNING	Wrong: send startup page activation, page view, and initial labels through three unrelated command bursts. Right: use one explicit startup sequence so the runtime transitions from cold start to the first meaningful frame atomically.

6.5 Real-time loop pattern at 60 Hz



60 Hz generated client loop with one batch per frame

For a classic external avionics client, the correct pattern is one loop at 60 Hz:

39. 1. sample external state
40. 2. mutate generated handles
41. 3. call `BuildCommandBatch(sequence)`
42. 4. publish the batch once
43. 5. sleep until the next tick

CODE CPP

```cpp
// Scenario: 60 Hz cockpit loop publishing HUD values, radar contacts, and status text.
using clock = std::chrono::steady_clock;
constexpr auto kTick = std::chrono::milliseconds(16);

std::uint32_t sequence = 1U;
auto nextTick = clock::now();

while (running)
{
    const auto frameStart = clock::now();                              // t = 0.0 ms
    const MissionSample sample = ReadMissionComputer();                // t = 0.2 ms

    PopulateHud(ui.Cockpit(), sample.hud);                             // t = 0.9 ms
    PopulateRadar(ui.Cockpit(), sample.radarTracks);                   // t = 1.8 ms
    ui.Cockpit().SetStatusCaption("WP-03 PUSH | THREAT SA-10");        // t = 2.0 ms

    const mfd::CommandBatch batch = ui.BuildCommandBatch(sequence); // t = 2.2 ms
    if (!client.SendBatch(batch))                                      // t = 2.5 ms
    {
        throw std::runtime_error("Unable to send frame batch: " + client.LastError());
```

```
    }

    ++sequence;
    nextTick += kTick;

    if (const auto now = clock::now(); now < nextTick)
    {
        std::this_thread::sleep_until(nextTick);                    // t = 16.6 ms target
    }
    else
    {
        nextTick = now;
    }
}
```

This is the reference pattern because the runtime processes a coherent per-frame command set. Scattering multiple send calls through the frame destroys sequencing, increases packet count, and makes feedback interpretation harder.

⚠ WARNING	Wrong: one `SendBatch()` for HUD text, another for page view, another for radar tracks. Right: one batch per client frame, one `sequence`, one coherent state transition.

Generated reset back to the authored state

Generated-root `Initialize()` is a deliberate user facility for "go back to the authored initial state". It is not the old per-cycle helper.

CODE CPP

```
// Scenario: return the runtime window to its authored baseline, then rebuild
// one fresh dynamic scene.
auto& staleTrack = ui.Cockpit().DynamicCockpitRadarContact().Create();
staleTrack.ContactLabel().SetText("OLD");
client.SendBatch(ui.BuildCommandBatch(10U));

ui.Initialize();
const bool staleHandleStillAlive = staleTrack.IsAlive(); // false

ui.Window().SetDisabled(true);
ui.Cockpit().SetStatusCaption("AUTHORED RESET");
auto& replacementTrack = ui.Cockpit().DynamicCockpitRadarContact().Create();
replacementTrack.ContactLabel().SetText("NEW");

if (!client.SendBatch(ui.BuildResetCommandBatch(11U)))
{
    throw std::runtime_error(client.LastError());
}
```

Behavior to rely on:

- `Initialize()` realigns the generated window, pages, static reticles, and strobes

to the authored baseline

- the next batch prepends one runtime `ResetWindowCommand`
- generated dynamic handles created before the reset become invalid and must be

recreated

- `IsAlive()` lets client code detect those stale dynamic handles explicitly

- `Run()` is the local per-cycle helper when no runtime reset is desired

■ 6.6 Page wrappers and authoritative active-page feedback

Each generated page wrapper exposes authored identity and authoritative active-page feedback:

- `Name()`
- `GeneratedId()`
- `MappingHash()`
- `IsActive()`

`IsActive()` is intentionally feedback-driven. It becomes true only when the runtime reports that the page is currently rendered as active. It does not become true merely because your client requested page activation.

This distinction matters for gating client logic. If your application should only publish page-specific overlays when the page is actually being rendered, gate that behavior on `IsActive()` rather than on your own last requested page variable.

■ 6.7 Window-level display controls

The UI root exposes `Window()` for whole-window state that is not tied to one page:

```cpp
// Scenario: dim the entire window for a night attack profile.
ui.Window().SetColorInverted(false);
ui.Window().SetBrightness(0.42f);
ui.Window().SetDisabled(false);

if (!client.SendBatch(ui.BuildCommandBatch(104U)))
{
    throw std::runtime_error(client.LastError());
}
```

Use this layer for whole-window presentation state such as brightness, inversion, or blackout. Do not overload reticle-level color changes when the real operational intent is a window-wide display mode change.

■ 6.8 Static reticle wrappers for persistent avionics values

Static reticle wrappers are the right surface for authored elements that always exist on the page and only need live value updates.

```cpp
// Scenario: update persistent HUD boxes with real mission values.
auto& cockpit = ui.Cockpit();

cockpit.hudSpeedBox.SetValue("420");
cockpit.hudMachBox.SetValue("0.64");
cockpit.hudHeadingBox.SetValue("045");
cockpit.hudFpaBox.SetValue("+02.5");
cockpit.hudThrottleBox.SetValue("88%");
cockpit.hudRadarBox.SetValue("EMIT");

if (overspeed)
{
```

```
    cockpit.hudSpeedBox.Blink = cockpit.overspeed;
    cockpit.hudMachBox.Blink = cockpit.overspeed;
}
else
{
    cockpit.hudSpeedBox.Blink = nullptr;
    cockpit.hudMachBox.Blink = nullptr;
}
```

This pattern keeps authored structure static while letting the client update operational values every frame.

⚠ WARNING	Wrong: rebuild the authored HUD layout as dynamic reticles just to update speed, heading, or throttle text. Right: keep persistent authored elements static and use the generated static wrappers as the live control surface.

Common reticle surface

Method	Use
SetVisible(bool)	show or hide the reticle
SetBlinkEnabled(bool)	toggle blinking without changing the blink type
SetBlinkType(const BlinkType&)	use one explicit page blink type
ClearBlinkType()	fall back to the page default blink behavior
SetPosition(Vec2)	move the whole reticle
SetRotationDegrees(float)	rotate the whole reticle
SetColor(ColorRgba)	override stroke color
SetThickness(float)	override line thickness

■ 6.9 Primitive handles for exposed geometry and text

Primitive handles are what make the generated API feel like an integration API instead of a string-based patch list. They let the client touch only the authored surface that was explicitly meant to move.

Normal generated code should therefore prefer:

- generated primitive accessors such as `HeadingValue()`, `TrackLabel()`, or

`FillBar()`

- generated convenience methods such as `SetValue(...)` when the wrapper emits

them

CODE CPP

```
// Scenario: steer the heading command bug and bank pointer from flight guidance.
auto& cockpit = ui.Cockpit();
```

```
cockpit.adiHeadingCommandBug.CommandBug().SetRotationDegrees(+27.0f);
cockpit.adiBankPointer.Pointer().SetRotationDegrees(-12.0f);
cockpit.adiHeadingBox.CommandValue().SetText("072");
cockpit.adiHeadingBox.HeadingValue().SetText("045");
```

Use primitive handles when the client needs one sub-element to move independently from the reticle that owns it. That is typical for command bugs, cue lines, text fields, and dynamic geometry markers.

The same rule applies to authored strobe reticles. If a strobe cursor contains an exposed text or line primitive, the generated page exposes that primitive through the authored strobe-reticle wrapper, and only the wrapper matching the currently selected strobe contributes commands.

Common primitive surface

Method	Use
SetVisible(bool)	toggle only this primitive
SetPosition(Vec2)	local translation inside the reticle
SetRotationDegrees(float)	local rotation
SetScale(Vec2)	local scale
SetColor(ColorRgba)	stroke color
SetThickness(float)	line thickness
SetLineStyle(LineStyle)	line style override

Fill-capable primitive surface

Only fill-capable generated handles additionally expose:

Method	Use
SetFillColor(ColorRgba)	fill color
SetFilled(bool)	fill toggle

These mutators exist on `CircleHandle`, `RingHandle`, `RectangleHandle`, `EllipseHandle`, `SquareHandle`, `DiamondHandle`, `TriangleHandle`, `PolylineHandle`, and `ArcHandle`. They do not exist on `TextHandle`, `TimeHandle`, `LineHandle`, `BezierHandle`, or `ImageHandle`. For polylines, fill is meaningful only when `SetClosed(true)` is also active.

Specialized surfaces

Handle	Extra methods
TextHandle	SetText, SetLetterSpacing
TimeHandle	SetLetterSpacing, SetTimeValue, ClearTimeValue, SetUtc, SetFieldVisibility

LineHandle	SetStart, SetEnd
CircleHandle	SetRadius
RingHandle	SetInnerRadius, SetOuterRadius, SetSegments
RectangleHandle	SetWidth, SetHeight, SetSize
EllipseHandle	SetWidth, SetHeight, SetSize
SquareHandle	SetWidth, SetHeight, SetSize
DiamondHandle	SetWidth, SetHeight, SetSize
TriangleHandle	SetPoints
PolylineHandle	SetPoints, SetClosed
BezierHandle	SetControlPoints, SetSegments
ArcHandle	SetRadius, SetStartAngleDegrees, SetEndAngleDegrees, SetSegments
ImageHandle	common primitive surface only

`TimeHandle` deliberately uses structured runtime fields. `SetTimeValue` provides the numeric calendar and clock value, `ClearTimeValue` stops that numeric bypass, `SetUtc` switches UTC/local rendering, and `SetFieldVisibility` selects which date/time fields are displayed. No raw runtime text format is accepted through the generated API.

⚠ WARNING	Wrong: expose every decorative primitive and then patch them from business code. Right: expose only the sub-elements that correspond to a real runtime control requirement.

■ 6.10 Dynamic reticles for radar tracks, threats, and steering cues

Dynamic reticles are the right model for runtime entities that appear, move, and disappear independently from the static page layout.

CODE CPP

```cpp
// Scenario: publish two live radar contacts and one threat cue on the cockpit page.
auto& cockpit = ui.Cockpit();
auto& contacts = cockpit.DynamicCockpitRadarContact();

auto& lead = contacts.Create();
lead.SetVisible(true);
lead.SetPosition({1.18f, 0.17f});
lead.SetRotationDegrees(-18.0f);
lead.SetColor({86, 244, 162, 255});
lead.ContactLabel().SetText("BRAA 045/32");
```

```
auto& threat = contacts.Create();
threat.SetVisible(true);
threat.SetPosition({0.92f, -0.08f});
threat.SetRotationDegrees(+36.0f);
threat.SetColor({255, 198, 109, 255});
threat.ContactLabel().SetText("THREAT SA-10");
threat.Blink = cockpit.threat;

cockpit.radarStatusBox.SetValue("THREAT SA-10");
cockpit.SetStatusCaption("WP-03 PUSH | COMMIT NORTH");
```

To remove one dynamic instance, remove the handle from the generated set and let the next batch carry the removal command:

CODE CPP

```
// Scenario: delete a stale contact when the track drops from the tactical picture.
contacts.Remove(threat);
```

The generated set hides three pieces of bookkeeping that user code should not own:

- the runtime reticle identifier
- the template transport id
- the `mappingHash` carried by the final batch

⚠ WARNING	Wrong: invent your own runtime-reticle-id scheme outside the generated set. Right: let `Create()` and `Remove()` manage lifecycle commands, then publish the resulting batch once per frame.

6.11 Build one coherent batch per frame

The two generated batch builders exist for different levels of transport control:

- `BuildBatch()` returns raw `std::vector<mfd::UserCommand>`
- `BuildResetBatch()` returns a runtime reset batch on the same generated surface
- `BuildCommandBatch(sequence)` returns one `mfd::CommandBatch` with both `sequence` and `mappingHash`
- `BuildResetCommandBatch(sequence)` does the same for a reset-oriented transaction

For production code, prefer the second form:

CODE CPP

```
// Scenario: publish one frame-aligned command batch with sequence tracking.
const mfd::CommandBatch batch = ui.BuildCommandBatch(sequence);
if (!client.SendBatch(batch))
{
    throw std::runtime_error(client.LastError());
}
```

Use `sequence` when the client operates in explicit update cycles. The runtime uses `sequence` together with `mappingHash` to reject stale or duplicate batches.

6.12 When to prefer LatestBatchPublisher

Use `LatestBatchPublisher` when the client continuously computes the latest state and only the freshest unsent frame matters.

Typical fit:

- radar sweeps
- HUD state streams
- synthetic moving maps
- continuously refreshed flight-symbology clients

Poor fit:

- low-rate command tools where every transaction must be delivered
- one-shot administrative commands
- flows where dropping intermediate pending frames is unacceptable

CODE CPP

```cpp
// Scenario: stream the freshest cockpit frame without blocking the producer loop.
mfd::client::LatestBatchPublisher publisher(*loaded.window.commandTransports.udp);
if (!publisher.IsReady())
{
    throw std::runtime_error("Unable to create realtime publisher: " +
publisher.LastError());
}

if (!ui.SubmitLatest(publisher, sequence))
{
    throw std::runtime_error("Unable to queue latest frame: " + publisher.LastError());
}
```

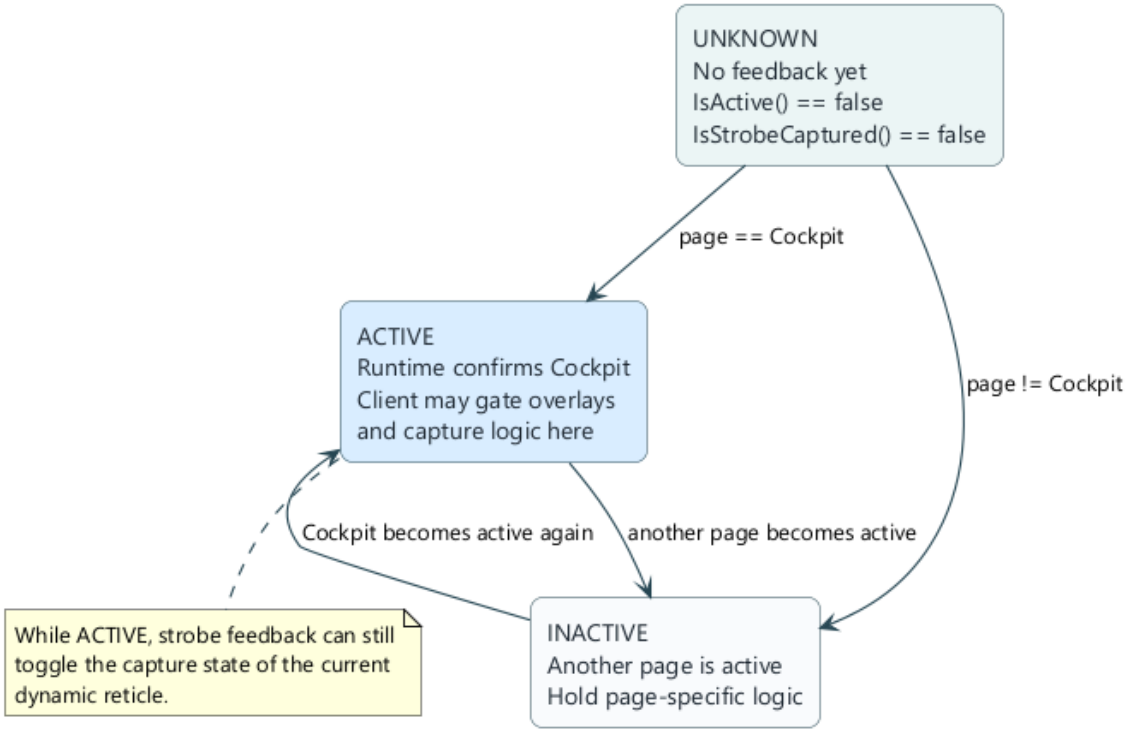
Implementation-backed behavior to know:

- one dedicated worker thread serializes sends
- `SubmitLatest()` replaces any older pending unsent batch with the newest one
- dynamic reticle lifecycle commands are preserved across pending-batch replacement when the `mappingHash` stays the same
- a pending `ResetWindowCommand` is preserved when a newer same-hash batch replaces an older unsent batch
- a newer reset drops older pending dynamic lifecycle commands because the runtime reset already supersedes that state
- `Flush()` blocks until the current send completes and no pending batch remains
- `Stop()` drops any unsent pending batch and terminates the worker thread

This makes `LatestBatchPublisher` a state-stream helper, not a guaranteed delivery queue.

⚠ WARNING	Wrong: use `LatestBatchPublisher` for every command path in the application. Right: reserve it for high-rate "latest state wins" publishing, and keep plain `SendBatch()` for explicit control transactions.

6.13 Feedback-driven state machine



Feedback-driven page activity and capture state machine

`IsActive()` and `IsStrobeCaptured()` should gate client behavior, not just feed telemetry logs.

CODE CPP

```
// Scenario: only commit the intercept logic when the runtime confirms both page activity
and strobe capture.
std::string feedbackError;
if (feedbackChannel)
{
    ui.PollFeedback(*feedbackChannel, 8U, &feedbackError);
}

auto& cockpit = ui.Cockpit();

if (!cockpit.IsActive())
{
    HoldRadarOverlay();
    return;
}

if (track != nullptr && track->IsStrobeCaptured())
{
    AuthorizeCommit("THREAT SA-10");
    cockpit.SetStatusCaption("WP-03 PUSH | CAPTURE CONFIRMED");
}
else
{
    ContinueSearch();
    cockpit.SetStatusCaption("WP-03 PUSH | SEARCHING");
}
```

Both signals are authoritative runtime feedback:

- `IsActive()` means the runtime is actually rendering that page
- `IsStrobeCaptured()` means the runtime confirmed that exact dynamic instance

as the captured target on the currently active page strobe

- `SetStrobeMagnetEnabled(bool)` remains a separate set-level publication

choice: it controls whether one dynamic template family may magnetize the strobe, but it does not suppress capture feedback for that family

⚠ WARNING	Wrong: assume capture as soon as the client places the strobe on top of a track. Right: wait for the runtime feedback path to confirm the active page and the captured runtime reticle, and treat magnet eligibility as a separate opt-in.

■ 6.14 What feedback guarantees

The generated feedback helpers intentionally expose only authoritative runtime state:

- `Page::IsActive()` is false until `ActivePageFeedback` is received
- `DynamicReticle::IsStrobeCaptured()` is false until the strobe feedback for

the currently active page identifies that exact runtime reticle

- `DynamicReticle::IsStrobeCaptured()` becomes false again when another page

becomes active until a fresh strobe snapshot arrives for the new active page

- `GeneratedDynamicReticleSet::SetStrobeMagnetEnabled(false)` keeps the set

non-magnetized but does not prevent the runtime from reporting capture when the strobe is positioned over one of its dynamic reticles

- stale feedback sequences are ignored by the runtime-feedback tracker
- if the feedback channel is disabled, these helpers remain conservative rather than speculative

This makes the generated feedback surface safe to use in client state machines.

Window shutdown detection is a separate, window-level feedback covered in section 6.20.

■ 6.15 Client sanitization and hygiene

The generated layer makes publication easier, but it does not sanitize domain data for you. A robust client still needs to validate its own upstream inputs.

Recommended safeguards:

- clamp logical coordinates to the authored range you accept
- reject or normalize non-finite floats before touching generated handles
- clamp brightness to `[0, 1]`
- cap loop delta time to avoid huge catch-up steps after a debugger stop
- keep one clear owner for dynamic-reticle lifecycle decisions
- surface `client.LastError()` and `publisher.LastError()` in operational logs

If your upstream data can contain `NaN`, `Inf`, or impossible kinematic values, sanitize before mutating the generated state tree.

6.16 When the raw API is still appropriate

The generated layer should be the default for window-specific integrations, but the raw `CommandClient` helpers still have a place:

- generic tooling that targets several windows
- low-level debugging tools
- migration code that predates the generated client surface

CODE CPP

```
// Scenario: a generic tactical tool injects one named dynamic reticle without including the generated header.
mfd::ReticlePatch patch;
patch.visible = true;
patch.position = mfd::Vec2 {0.24f, -0.11f};
patch.text = std::string {"BRAA 045/32"};

if (!client.UpsertDynamicReticle("Cockpit", "lead_track", "cockpit_radar_contact", patch))
{
    throw std::runtime_error(client.LastError());
}
```

Use the raw path for genericity, not as a substitute for the generated path in a window-specific product client.

6.17 When and how to use UserSpaceProjector

`UserSpaceProjector` is the right tool when your client does not naturally work in authored page coordinates.

Typical cases:

- one moving local frame centered on an aircraft, vehicle, ship, or sensor
- domain positions expressed in nautical miles, meters, or another physical unit
- headings expressed in radians
- one page that must stay visually aircraft-centric while the ownship moves and turns

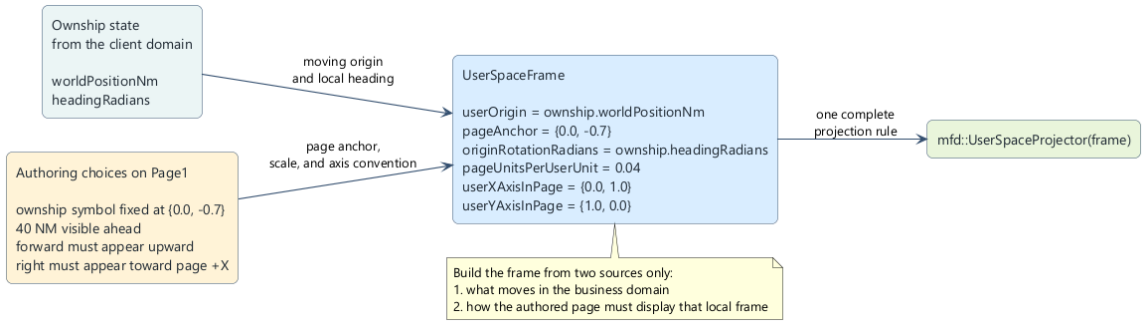
Do not use it just because the helper exists. If your client already computes positions directly in page space `[-1, 1]` and rotations directly in page-space degrees, sending those values without a projector is simpler and clearer.

Situation	Use UserSpaceProjector?	Why
tracks are already computed in page coordinates	no	no extra frame conversion is needed
one authored page must stay centered on one moving aircraft	yes	the helper isolates origin, axis, scale, and rotation conversion
the client owns positions in NM and azimuth/range around the aircraft	yes	keep domain units client-side until the final page-space write
one static decoration always	no	author it directly in JSON and

stays at one fixed authored position		leave it static

⬡ API	`UserSpaceProjector` is client-side only. The runtime still receives ordinary page-space coordinates in `[-1, 1]` and ordinary rotations in degrees.

6.17.1 Build the projector from the page contract and the business frame



How the `client_tutorial` builds `UserSpaceFrame` for `Page1`

You should build `UserSpaceProjector` from two independent decisions:

44. 1. the business frame owned by the client
45. 2. the authored page contract that tells you where and how that frame must appear on the page

For the `client_tutorial`, the business frame is:

- ownship world position in nautical miles
- ownship heading in radians
- track states stored as `distanceNm`, `azimuthRadians`, and `relativeHeadingRadians`

The authored page contract for `Page1` is:

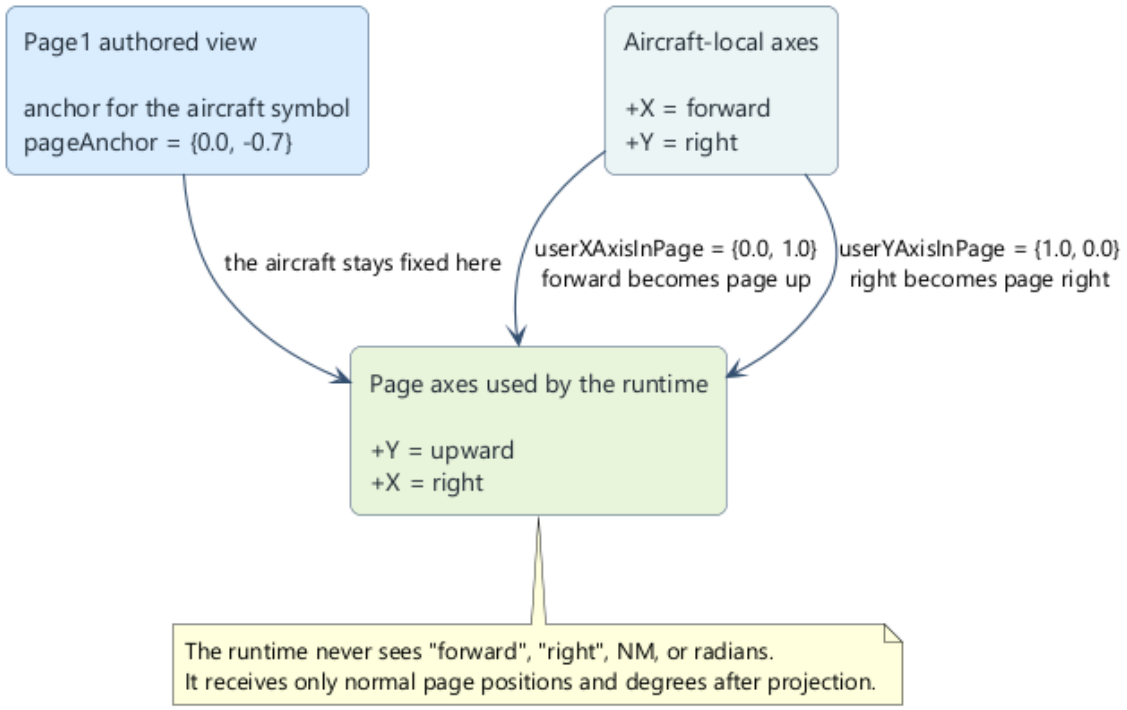
- the ownship symbol is drawn once and stays fixed at `{0.0f, -0.7f}`
- the farthest straight-ahead contact must remain inside the page
- aircraft forward must appear upward on the page
- aircraft right must appear toward page `+X`

That directly gives the six `UserSpaceFrame` fields:

UserSpaceFrame field	client_tutorial value	Why
userOrigin	ownship.worldPositionNm	the moving frame is centered on the aircraft
pageAnchor	{0.0f, -0.7f}	the authored aircraft symbol stays fixed there
originRotationRadians	ownship.headingRadians	the local frame rotates with the aircraft
pageUnitsPerUserUnit	0.04f	40 NM must fit in the forward page span

userXAxisInPage	{0.0f, 1.0f}	aircraft +X means "forward", rendered upward
userYAxisInPage	{1.0f, 0.0f}	aircraft +Y means "right", rendered toward page +X

6.17.2 The exact Page1 frame used by client_tutorial



Axis mapping from aircraft-local coordinates to the authored `Page1`

The implementation in `examples/client_tutorial/src/main.cpp` is deliberately small because all the meaning lives in the six frame fields:

CODE CPP

```
mfd::UserSpaceProjector BuildTutorialProjector(const OwnshipState& ownship)
{
    mfd::UserSpaceFrame frame;
    frame.userOrigin = ownship.worldPositionNm;
    frame.pageAnchor = {0.0f, -0.7f};
    frame.originRotationRadians = ownship.headingRadians;
    frame.pageUnitsPerUserUnit = (0.9f - (-0.7f)) / 40.0f;
    frame.userXAxisInPage = {0.0f, 1.0f};
    frame.userYAxisInPage = {1.0f, 0.0f};
    return mfd::UserSpaceProjector(frame);
}
```

Read it in this order:

46. 1. `userOrigin`: "what point in my domain frame is the moving origin?"
47. 2. `pageAnchor`: "where must that origin appear on the authored page?"
48. 3. `originRotationRadians`: "what local heading must be considered page forward?"
49. 4. `pageUnitsPerUserUnit`: "how much page space does one NM represent?"
50. 5. `userXAxisInPage` and `userYAxisInPage`: "how do aircraft-local axes map to page axes?"

The scale deserves one explicit derivation:

CODE TEXT

```
forward visible page span = 0.9 - (-0.7) = 1.6 page units
maximum forward range     = 40 NM
pageUnitsPerUserUnit      = 1.6 / 40 = 0.04
```

That means:

- `1 NM` becomes `0.04` page units
- `10 NM` becomes `0.40` page units
- `20 NM` becomes `0.80` page units before anchor offset is added

6.17.3 Detailed worked example with range and azimuth around the aircraft

From aircraft-relative polar data to generated page-space commands

The important part of `client_tutorial` is not the final `SetPosition(...)` call. The important part is that the source state stays aircraft-centric until the last moment.

One track is stored like this:

CODE CPP

```
struct TutorialTrackState
{
    float distanceNm = 20.0f;
    float azimuthRadians = 0.40f;
    float relativeHeadingRadians = -0.20f;
};
```

Interpretation:

- `distanceNm = 20.0f`: the target is `20 NM` from the aircraft
- `azimuthRadians = 0.40f`: the target is offset by `0.40 rad` around the aircraft nose
- `relativeHeadingRadians = -0.20f`: the target heading is `0.20 rad` left of aircraft forward

The client first converts range/azimuth to one aircraft-local cartesian offset:

CODE TEXT

```
localOffsetNm.x = distanceNm * cos(azimuthRadians) = 20 * cos(0.40) = 18.42
localOffsetNm.y = distanceNm * sin(azimuthRadians) = 20 * sin(0.40) =  7.79
```

With the `Page1` axis mapping:

- aircraft-local `+X` becomes page up
- aircraft-local `+Y` becomes page right

So before anchor offset:

CODE TEXT

```
pageOffset = { localOffsetNm.y, localOffsetNm.x } * 0.04
           = { 7.79, 18.42 } * 0.04
           = { 0.31, 0.74 }
```

Then the aircraft anchor is applied:

CODE TEXT

```
pagePosition = pageAnchor + pageOffset
             = { 0.0, -0.7 } + { 0.31, 0.74 }
             = { 0.31, 0.04 }
```

This is why the same target appears:

- slightly to the right of the page center

- above the aircraft anchor
- without ever storing raw page coordinates in the business state

The code path used by `client_tutorial` follows that same logic:

```cpp
const mfd::Vec2 localOffsetNm =
    BuildOwnshipRelativeOffsetNm(track.azimuthRadians, track.distanceNm);
const mfd::Vec2 worldPositionNm =
    ownship.worldPositionNm + RotateRadians(localOffsetNm, ownship.headingRadians);

track.reticle->SetPosition(projector.ToPagePosition(worldPositionNm));
track.reticle->SetRotationDegrees(
    projector.ToPageRotationDegrees(
        ownship.headingRadians + track.relativeHeadingRadians));
```

Why is the projector still useful if the track was rebuilt as one world position first? Because the same projector:

- subtracts the moving aircraft origin
- removes the ownship heading from the projected result
- reapplies the authored page axis convention
- keeps the page aircraft-centric even while the aircraft moves and turns

6.17.4 Update rule for a live loop

In an aircraft-centric page, rebuild or refresh the frame every cycle before you project the tracks:

```cpp
mfd::UserSpaceFrame frame = projector.Frame();
frame.userOrigin = ownship.worldPositionNm;
frame.originRotationRadians = ownship.headingRadians;
projector.SetFrame(frame);
```

Then:

51. 1. keep track state in NM and radians
52. 2. project positions only when filling generated handles
53. 3. send the final batch in normal page space

This rule matters because the projector is not a scene graph. It is only one pure client-side conversion step. If the aircraft moved or turned, the frame must reflect that before the next projection.

⚠ WARNING	Do not author the page in nautical miles. Author the page normally in `[-1, 1]`, keep the aircraft symbol fixed where you want it, and convert only in the client.

6.17.5 When not to use UserSpaceProjector

Skip the projector when:

- the client already owns stable page-space coordinates
- one page is not centered on a moving local frame
- you are updating only static text or simple authored widgets
- adding the helper would only duplicate one trivial direct `SetPosition(...)`

The helper is valuable only when it removes coordinate-system coupling from the rest of the client code.

6.18 Minimal end-to-end example

CODE CPP

```cpp
// Scenario: minimum typed client that activates the cockpit page and publishes one radar-ready frame.
#include "MockupUi.h"
#include "mfd/client/ClientSdk.h"

int main()
{
    mfd::JsonLoader loader;
    const mfd::LoadedWindowConfiguration loaded =

loader.LoadWindowConfiguration(std::string(mockup_ui::CockpitMockupUi::WindowFile()));

    if (!loaded.window.commandTransports.udp.has_value() ||
        !loaded.generatedTransportMap.has_value())
    {
        return 1;
    }

    mfd::CommandClient client(
        *loaded.window.commandTransports.udp,
        loaded.generatedTransportMap);
    if (!client.IsReady())
    {
        return 1;
    }

    mockup_ui::CockpitMockupUi ui;
    auto& cockpit = ui.Cockpit();

    if (!client.ActivatePage(cockpit))
    {
        return 1;
    }

    ui.Window().SetBrightness(0.65f);
    cockpit.hudHeadingBox.SetValue("045");
    cockpit.radarStatusBox.SetValue("SEARCH");
    cockpit.SetStatusCaption("WP-03 READY | BRAA 045/32");

    return client.SendBatch(ui.BuildCommandBatch(1U)) ? 0 : 1;
}
```

6.19 Render a window offscreen with mfd_runtime_api

Use `mfd_runtime_api` when your application must load one authored window, keep the authored UDP in/out contract, and render the runtime into one application-owned image without launching `mfd_window`.

This package is the dedicated public surface for that use case:

- it exposes `RuntimeSession` and `OffscreenSurface`
- it keeps `mfd_api` hidden from the consumer-facing API
- it keeps low-level render dependencies private at the package boundary
- it lets the host application decide how to present the produced pixels

The intended separation is simple:

- `mfd_window` + framebuffer plugin: when you want the generic runtime host and one DLL capture hook
- `mfd_runtime_api`: when you want to embed the runtime directly inside your own application

6.19.1 Consumption contract

From an external CMake project, consume the package directly:

CODE CMAKE

```
find_package(MFDStudioRuntimeApi CONFIG REQUIRED)
target_link_libraries(my_host_app PRIVATE MFDStudio::RuntimeApi)
```

The host application may use any presentation layer it wants for the final image. If it needs a GUI or a GPU upload path, that dependency stays the host application's choice rather than part of the `mfd_runtime_api` public contract.

6.19.2 Runtime objects and responsibilities

`RuntimeSession` owns the authored runtime state:

- load one window JSON
- keep the authored UDP command and feedback configuration
- advance the runtime frame
- switch pages and inspect lightweight status

`OffscreenSurface` owns one independent offscreen output:

- explicit `Resize(width, height)` controlled by the host application
- one private render target per surface
- one private CPU readback buffer per surface
- one `FrameView()` for the latest rendered image

Because each surface owns its own target and CPU buffer, multiple windows or multiple views can be rendered in the same host frame without texture collisions.

6.19.3 Clipping and resizing guarantees

The offscreen path preserves the same clipping semantics as the window path. Each `OffscreenSurface` uses its own stencil-capable target, so reticle clipping remains valid after resize and does not leak from one surface to another.

The host application is responsible for calling `Resize(...)` whenever its own presentation area changes size. This keeps the offscreen output aligned with the exact dimensions requested by the application instead of an implicit window size.

6.19.4 Minimal host loop

CODE CPP

```
// Scenario: host one authored window offscreen and upload the latest image elsewhere.
#include "mfd/runtime_api/OffscreenSurface.h"
#include "mfd/runtime_api/RuntimeSession.h"

mfd::runtime_api::RuntimeSession runtimeSession;
std::string error;

if (!runtimeSession.LoadWindowFile("assets/windows/demo_pages_minimal.json", error))
{
    return 1;
}

mfd::runtime_api::OffscreenSurface surface(800, 600);

for (;;)
{
    runtimeSession.Advance(dtSeconds);
    surface.Resize(currentWidth, currentHeight);
    surface.Render(runtimeSession);

    const mfd::runtime_api::OffscreenFrameView frame = surface.FrameView();
    UploadFrameToMyUi(frame.pixels, frame.width, frame.height, frame.rowStrideBytes);
}
```

The host loop stays intentionally small:

54. 1. advance the runtime
55. 2. resize the offscreen surface if the host view changed
56. 3. render the runtime
57. 4. upload or copy the returned frame into the host application's own UI layer

6.19.5 Reference example

The repository ships one complete example:

CODE TEXT

```
examples/offscreen_viewer/src/main.cpp
```

That example:

- does not use `WindowLauncher` scripts
- loads the runtime through `mfd_runtime_api`
- keeps UDP in/out active from the authored window
- renders two independent offscreen surfaces
- displays both uploaded images in a resizable host window

Build it with:

CODE POWERSHELL

```
cmake --build --preset debug-win32 --target offscreen_viewer
```

Use this path when your goal is embedding. Keep the framebuffer plugin ABI for the `mfd_window` host path only.

6.20 Detect window shutdown and reset the client

The command and feedback streams are connectionless UDP. A client cannot rely on a socket error to learn that the window vanished, so the runtime publishes a dedicated window lifecycle feedback. It is a window-level liveness signal, intentionally independent of any page or strobe:

- the window emits a periodic `Alive` heartbeat on the configured feedback

heartbeat cadence, even when no page or strobe is active

- the window emits one final `Closing` payload during a graceful shutdown, just

before it tears down its transports

On the client side, `mfd::client::WindowLivenessMonitor` turns those two signals into a single connection state. It combines the explicit `Closing` (immediate reaction) with an absence-of-heartbeat timeout (covers crash, kill, or network loss). The monitor keeps no internal clock: you pass a monotonic time in seconds, which keeps it deterministic and testable.

CODE CPP

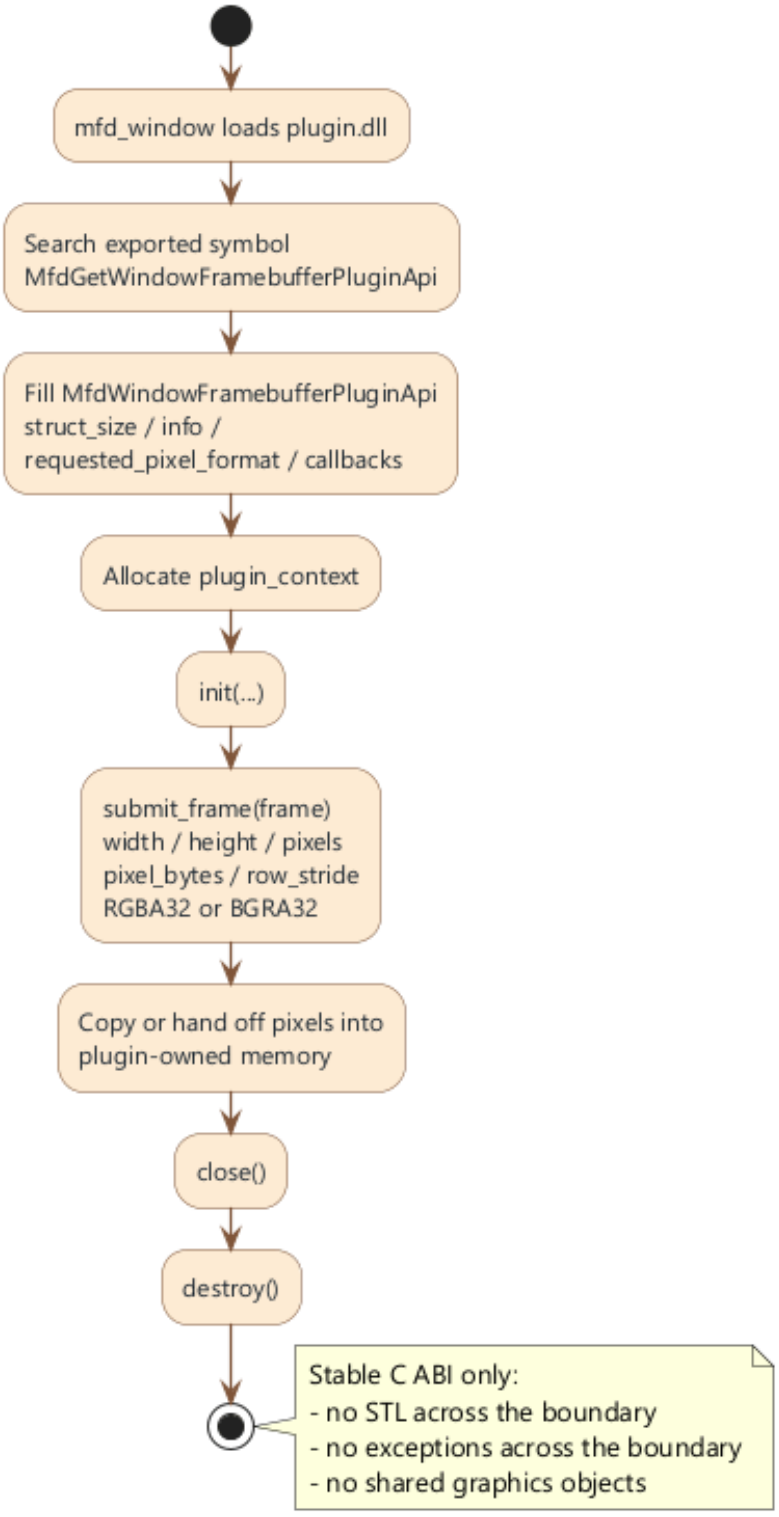
```
mfd::client::WindowLivenessMonitor liveness(2.0); // timeout > heartbeat interval

// Once per frame, after polling feedback through the generated UI:
ui.PollFeedback(*feedbackChannel, 8U, &feedbackError);
liveness.Observe(ui.TotalDecodedFeedbackPackets(),
                 ui.WindowReportedClosing(),
                 nowSeconds);

if (liveness.ConsumeDisconnect())
{
    // The window closed or stopped answering: drop the stale transports and
    // rebuild them so a relaunched window starts from a clean baseline.
    client = mfd::CommandClient(*loaded.window.commandTransports.udp,
loaded.generatedTransportMap);
    feedbackChannel =
mfd::CreateFeedbackReceiverChannel(*loaded.window.feedbackTransports.udp);
    ui.Initialize();       // re-send the full authored state on the next batch
    liveness.Reset();      // re-arm for the next connection
}
```

Clients that decode the raw feedback channel themselves (instead of the generated `PollFeedback`) feed the monitor directly: `RecordActivity(nowSeconds)` for every decoded packet, `RecordWindowClosing(nowSeconds)` when a `Closing` payload arrives, and `Update(nowSeconds)` once per frame to apply the timeout.

Choose a timeout safely larger than the window feedback heartbeat interval so a single dropped UDP heartbeat does not raise a false disconnect. The detection only works against a window that exposes a feedback transport: a command-only window emits no heartbeat and cannot be monitored.

The shipped examples illustrate both ends of the pattern: `client_tutorial` and `client_mockup` rebuild their transports and reconnect, while `client_mockup_minimal` detects the shutdown and stops cleanly.

7. Write a framebuffer capture DLL plugin



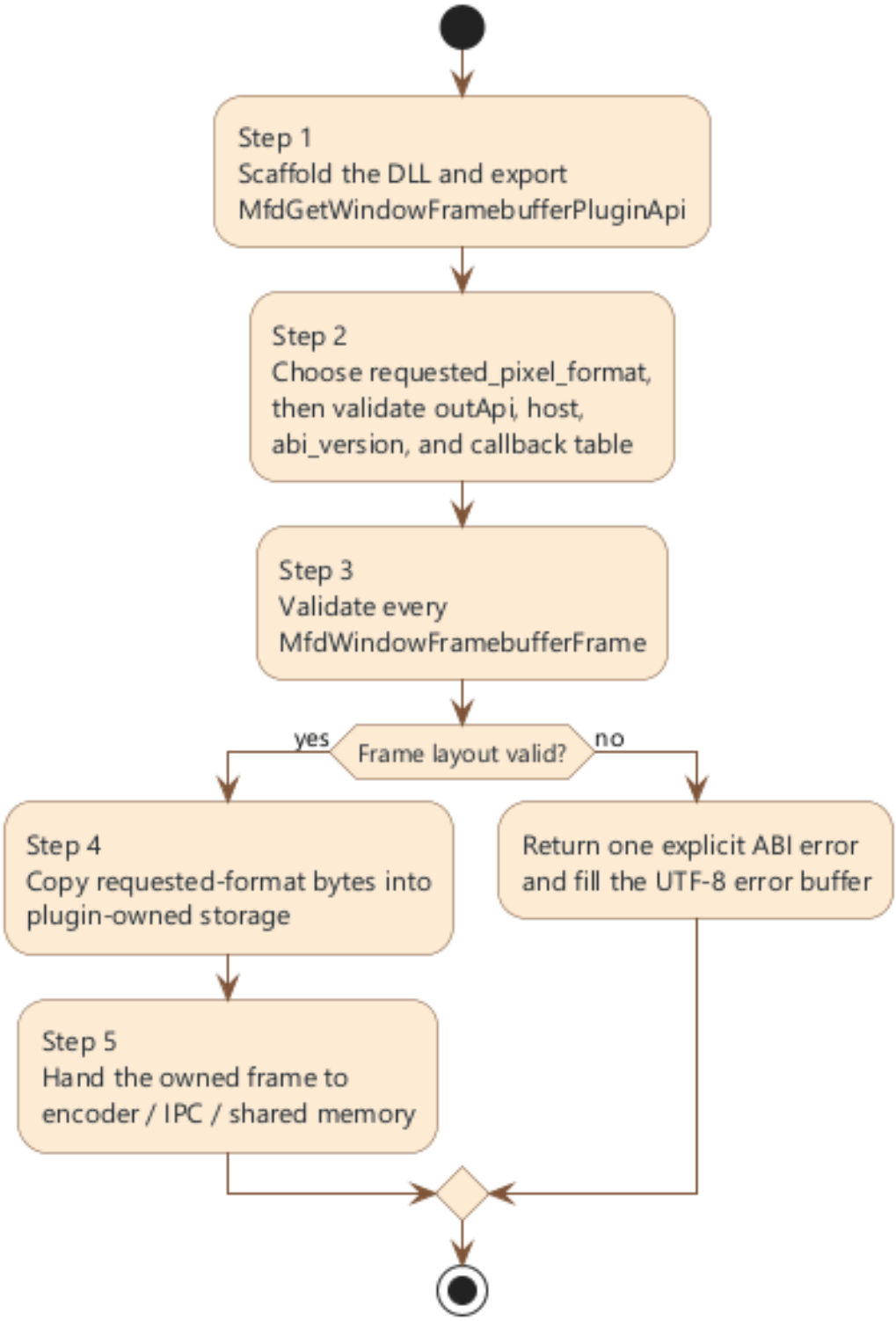
ABI framebuffer plugin

This chapter is intentionally written as a standalone integration guide. You should be able to implement, validate, and test a framebuffer plugin without reading the rest of the manual first.

Recommended order:

58. 1. scaffold the plugin and export the stable entry point

59. 2. request the output pixel format and validate the ABI contract in the factory and `init`
60. 3. validate every frame descriptor
61. 4. copy or hand off the pixels before `submit_frame` returns
62. 5. connect the async encoder or IPC path behind that handoff



Framebuffer plugin integration workflow for an application team

7.1 Why the framebuffer plugin ABI is C-only

The plugin ABI intentionally avoids C++ implementation details across the DLL boundary:

- no STL containers
- no C++ class layout assumptions
- no exceptions across the boundary
- no host-owned graphics objects
- no dependency on one compiler-specific ABI

The public header is:

```text
mfd_window_plugin_api/include/mfd/window/WindowLauncherPlugin.h
```

If you keep the DLL boundary C-only, you can freely choose your internal C++17 implementation behind that boundary.

7.2 Step 1: scaffold the plugin and export the entry point

The runtime looks for one exported symbol:

```cpp
MfdGetWindowFramebufferPluginApi
```

The public header also exposes the same name through:

```cpp
MFD_WINDOW_FRAMEBUFFER_PLUGIN_ENTRY_POINT
```

Your first deliverable is a DLL that exports that factory symbol and fills one `MfdWindowFramebufferPluginApi` structure.

7.3 Step 2: validate ABI and host contract during startup

`MfdWindowFramebufferPluginApi` is the exported callback table:

Field	Why it matters
struct_size	forward-compatibility and layout validation
info	immutable plugin metadata
plugin_context	plugin-owned opaque state
init	host-to-plugin startup callback
submit_frame	per-frame entry point
close	shutdown callback before unload
destroy	final context destruction hook

`MfdWindowFramebufferPluginInfo` describes the plugin instance:

Field	Use
struct_size	versioned metadata layout
abi_version	expected ABI revision
plugin_id	stable machine-readable plugin identifier
display_name	human-readable plugin name
requested_pixel_format	output layout requested for submit_frame

At minimum, validate these conditions in the factory and `init`:

- `outApi != nullptr`
- `host != nullptr`
- `host->abi_version == MFD_WINDOW_FRAMEBUFFER_PLUGIN_ABI_VERSION`
- `host->output_pixel_format == outApi->info.requested_pixel_format`
- every mandatory callback is populated

7.4 Step 3: understand the frame descriptor you receive

`submit_frame` receives one `MfdWindowFramebufferFrame` borrowed from the host:

Field	Meaning
struct_size	versioned frame structure size
pixel_format	actual output layout selected by the host
width	frame width in pixels
height	frame height in pixels
row_stride_bytes	number of bytes between two rows
pixels	borrowed pointer to the first pixel byte
pixel_bytes	total byte count available through pixels

The stable ABI currently supports:

- `MfdWindowFramebufferPixelFormat_Rgba32`
- `MfdWindowFramebufferPixelFormat_Bgra32`, compatible with `GL_BGRA_EXT`

Assume only what the ABI states. Do not infer that GPU resources are shared, that the pixel memory survives beyond the callback, or that another format has the same channel order.

7.5 Step 4: validate every frame descriptor defensively

The ABI already exposes helpers for the most important validation steps:

Helper	Use
MfdWindowComputeFramebufferByteCount	expected byte count for one format / width / height tuple
MfdWindowValidateFramebufferLayout	validate raw tightly packed RGBA32 or BGRA32 bytes
MfdWindowComputeFramebufferRgba32ByteCount	expected byte count for one width/height pair
MfdWindowValidateFramebufferRgba32Layout	validate raw RGBA32 layout and byte count
MfdWindowValidateFramebufferFrame	validate the full frame descriptor

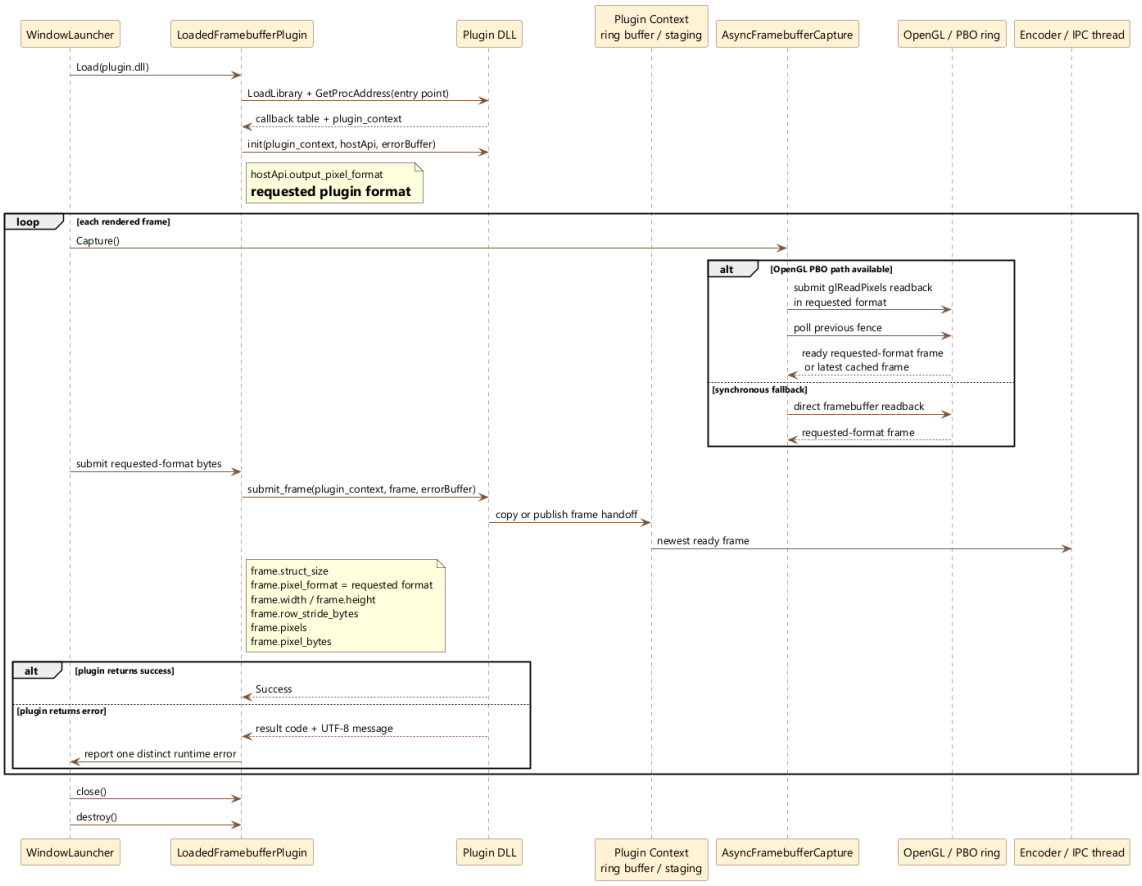
Use them at the start of `submit_frame`:

CODE CPP

```cpp
// Scenario: reject malformed frames before touching plugin-owned buffers.
MfdWindowFramebufferPluginResultCode MFD_WINDOW_PLUGIN_CALL SubmitFramePlugin(
    void* pluginContext,
    const MfdWindowFramebufferFrame* frame,
    MfdWindowUtf8Buffer* error) noexcept
{
    if (pluginContext == nullptr || frame == nullptr)
    {
        return MfdWindowFramebufferPluginResultCode_InvalidArgument;
    }

    if (MfdWindowValidateFramebufferFrame(frame) == 0)
    {
        return MfdWindowFramebufferPluginResultCode_InvalidArgument;
    }

    return MfdWindowFramebufferPluginResultCode_Success;
}
```

If validation fails, return one explicit ABI result code and write one human-readable error string into the provided UTF-8 buffer when possible.

7.6 Step 5: know the lifecycle before you attach real logic



Framebuffer capture runtime sequence and async frame handoff

The normal runtime lifecycle is:

63. 1. `mfd_window` loads the DLL
64. 2. the loader calls `MfdGetWindowFramebufferPluginApi`
65. 3. the plugin allocates its context and returns the callback table
66. 4. the runtime calls `init`
67. 5. the runtime calls `submit_frame` once per rendered frame
68. 6. the runtime calls `close`
69. 7. the runtime calls `destroy`

Design your context so that `init`, `submit_frame`, `close`, and `destroy` each have one clear responsibility. Do not let `submit_frame` lazily initialize half of the encoder stack unless you are forced to.

7.7 Minimal plugin scaffold

CODE CPP

```
// Scenario: minimal plugin factory that validates ABI and returns one callback table.
#include "mfd/window/WindowLauncherPlugin.h"

struct PluginContext
{
    bool initialized = false;
};

MfdWindowFramebufferPluginResultCode MFD_WINDOW_PLUGIN_CALL InitPlugin(
    void* pluginContext,
    const MfdWindowFramebufferPluginHostApi* host,
```

```
    MfdWindowUtf8Buffer* error) noexcept
{
    if (pluginContext == nullptr || host == nullptr ||
        host->abi_version != MFD_WINDOW_FRAMEBUFFER_PLUGIN_ABI_VERSION ||
        host->output_pixel_format != MfdWindowFramebufferPixelFormat_Bgra32)
    {
        return MfdWindowFramebufferPluginResultCode_InvalidArgument;
    }

    static_cast<PluginContext*>(pluginContext)->initialized = true;
    return MfdWindowFramebufferPluginResultCode_Success;
}

void MFD_WINDOW_PLUGIN_CALL ClosePlugin(void* pluginContext) noexcept
{
    (void)pluginContext;
}

void MFD_WINDOW_PLUGIN_CALL DestroyPlugin(void* pluginContext) noexcept
{
    delete static_cast<PluginContext*>(pluginContext);
}

extern "C" __declspec(dllexport) MfdWindowFramebufferPluginResultCode MFD_WINDOW_PLUGIN_CALL
MfdGetWindowFramebufferPluginApi(MfdWindowFramebufferPluginApi* outApi, MfdWindowUtf8Buffer*
error) noexcept
{
    if (outApi == nullptr)
    {
        return MfdWindowFramebufferPluginResultCode_InvalidArgument;
    }

    auto* context = new (std::nothrow) PluginContext();
    if (context == nullptr)
    {
        return MfdWindowFramebufferPluginResultCode_InternalFailure;
    }

    *outApi = {};
    outApi->struct_size = sizeof(*outApi);
    outApi->info.struct_size = sizeof(outApi->info);
    outApi->info.abi_version = MFD_WINDOW_FRAMEBUFFER_PLUGIN_ABI_VERSION;
    outApi->info.requested_pixel_format = MfdWindowFramebufferPixelFormat_Bgra32;
    outApi->plugin_context = context;
    outApi->init = &InitPlugin;
    outApi->submit_frame = &SubmitFramePlugin;
    outApi->close = &ClosePlugin;
    outApi->destroy = &DestroyPlugin;
    return MfdWindowFramebufferPluginResultCode_Success;
}
```

This is the correct starting point: a tiny stable C boundary, and all implementation detail hidden behind `plugin_context`.

■ 7.8 Copy pixels safely inside submit_frame

The runtime owns `frame->pixels`. Your plugin only borrows that memory during the callback.

CODE CPP

```
// Scenario: copy one plugin-selected raw frame into plugin-owned storage before returning.
struct PluginContext
{
    std::vector<std::uint8_t> staging;
    int width = 0;
    int height = 0;
    std::size_t stride = 0;
    std::uint32_t pixelFormat = 0;
};

MfdWindowFramebufferPluginResultCode MFD_WINDOW_PLUGIN_CALL SubmitFramePlugin(
    void* pluginContext,
    const MfdWindowFramebufferFrame* frame,
    MfdWindowUtf8Buffer* error) noexcept
{
    if (pluginContext == nullptr || frame == nullptr ||
        MfdWindowValidateFramebufferFrame(frame) == 0)
    {
        return MfdWindowFramebufferPluginResultCode_InvalidArgument;
    }

    auto& context = *static_cast<PluginContext*>(pluginContext);
    context.staging.resize(frame->pixel_bytes);
    std::memcpy(context.staging.data(), frame->pixels, frame->pixel_bytes);
    context.width = frame->width;
    context.height = frame->height;
    context.stride = frame->row_stride_bytes;
    context.pixelFormat = frame->pixel_format;
    return MfdWindowFramebufferPluginResultCode_Success;
}
```

✕ DANGER	Never keep `frame->pixels` after `submit_frame` returns. The pointer is borrowed host memory whose lifetime is only guaranteed for the duration of the callback.

7.9 Connect an async encoder or IPC path without blocking submit_frame

Once the safe copy rule is respected, the usual next step is to hand the copied frame to another thread.

CODE CPP

```
// Scenario: pseudocode for a lock-free ring buffer handoff to an encoder thread.
struct FrameSlot
{
    std::atomic<bool> ready {false};
    std::vector<std::uint8_t> pixels;
    int width = 0;
    int height = 0;
    std::size_t stride = 0;
};

struct PluginContext
{
    std::array<FrameSlot, 3> ring;
    std::atomic<std::uint32_t> writeIndex {0};
    std::atomic<std::uint32_t> readIndex {0};
};

MfdWindowFramebufferPluginResultCode SubmitFramePlugin(...) noexcept
```

```
{
    FrameSlot& slot = context.ring[context.writeIndex.load() % context.ring.size()];
    if (slot.ready.load(std::memory_order_acquire))
    {
        return MfdWindowFramebufferPluginResultCode_Success; // drop this frame instead of
blocking
    }

    slot.pixels.resize(frame->pixel_bytes);
    std::memcpy(slot.pixels.data(), frame->pixels, frame->pixel_bytes);
    slot.width = frame->width;
    slot.height = frame->height;
    slot.stride = frame->row_stride_bytes;
    slot.ready.store(true, std::memory_order_release);
    context.writeIndex.fetch_add(1, std::memory_order_relaxed);
    return MfdWindowFramebufferPluginResultCode_Success;
}

void EncoderThreadMain(PluginContext& context)
{
    for (;;)
    {
        FrameSlot& slot = context.ring[context.readIndex.load() % context.ring.size()];
        if (!slot.ready.load(std::memory_order_acquire))
        {
            continue;
        }

        EncodeOrPublish(slot.width, slot.height, slot.stride, slot.pixels);
        slot.ready.store(false, std::memory_order_release);
        context.readIndex.fetch_add(1, std::memory_order_relaxed);
    }
}
```

This pattern gives `submit_frame` a bounded job:

- validate the frame
- copy it into owned storage
- publish the ownership handoff
- return immediately

That is the right place to connect video encoding, shared memory publication, or IPC forwarding.

7.10 What not to do

Do not:

- keep `frame->pixels` beyond the callback
- throw exceptions across the ABI
- assume a pixel format other than the one validated by the helpers
- block the render loop on disk I/O, encoding, or network publication
- write into the memory exposed by `frame->pixels`

If your plugin must choose between dropping one frame and blocking the render loop, prefer the explicit drop strategy and expose it in diagnostics.

7.11 Test the plugin without the full runtime

You can unit-test most of the plugin contract without launching `mfd_window`. Build a fake frame on the stack and call your callbacks directly.

CODE CPP

```cpp
// Scenario: unit-test the plugin against one synthetic 2x2 BGRA32 frame.
std::array<std::uint8_t, 16> pixels {};

MfdWindowFramebufferFrame frame {};
frame.struct_size = sizeof(frame);
frame.pixel_format = MfdWindowFramebufferPixelFormat_Bgra32;
frame.width = 2;
frame.height = 2;
frame.row_stride_bytes = 8;
frame.pixels = pixels.data();
frame.pixel_bytes = pixels.size();

EXPECT_NE(MfdWindowValidateFramebufferFrame(&frame), 0);

PluginContext context;
MfdWindowUtf8Buffer error {};
EXPECT_EQ(
    SubmitFramePlugin(&context, &frame, &error),
    MfdWindowFramebufferPluginResultCode_Success);
EXPECT_EQ(context.staging.size(), pixels.size());
```

This isolates plugin logic from runtime launch, OpenGL capture, and DLL loading. It is the fastest way to prove:

- frame validation works
- copy logic works
- async queue handoff works
- error reporting works on malformed descriptors

7.12 Repository reference plugin

The repository already includes one minimal reference implementation:

CODE TEXT

```
examples/mfd_framebuffer_stdout_plugin/src/FramebufferStdoutPlugin.cpp
```

That sample proves:

- callback-table export
- explicit output-format request
- ABI validation
- safe frame validation
- context allocation and destruction

Start from it if you need a clean skeleton, then add your own queueing, encoder integration, or IPC path.

7.13 CMake and build notes

The reference plugin is a standard C++17 DLL. The critical points are not CMake tricks; they are ABI correctness and symbol export.

Checklist:

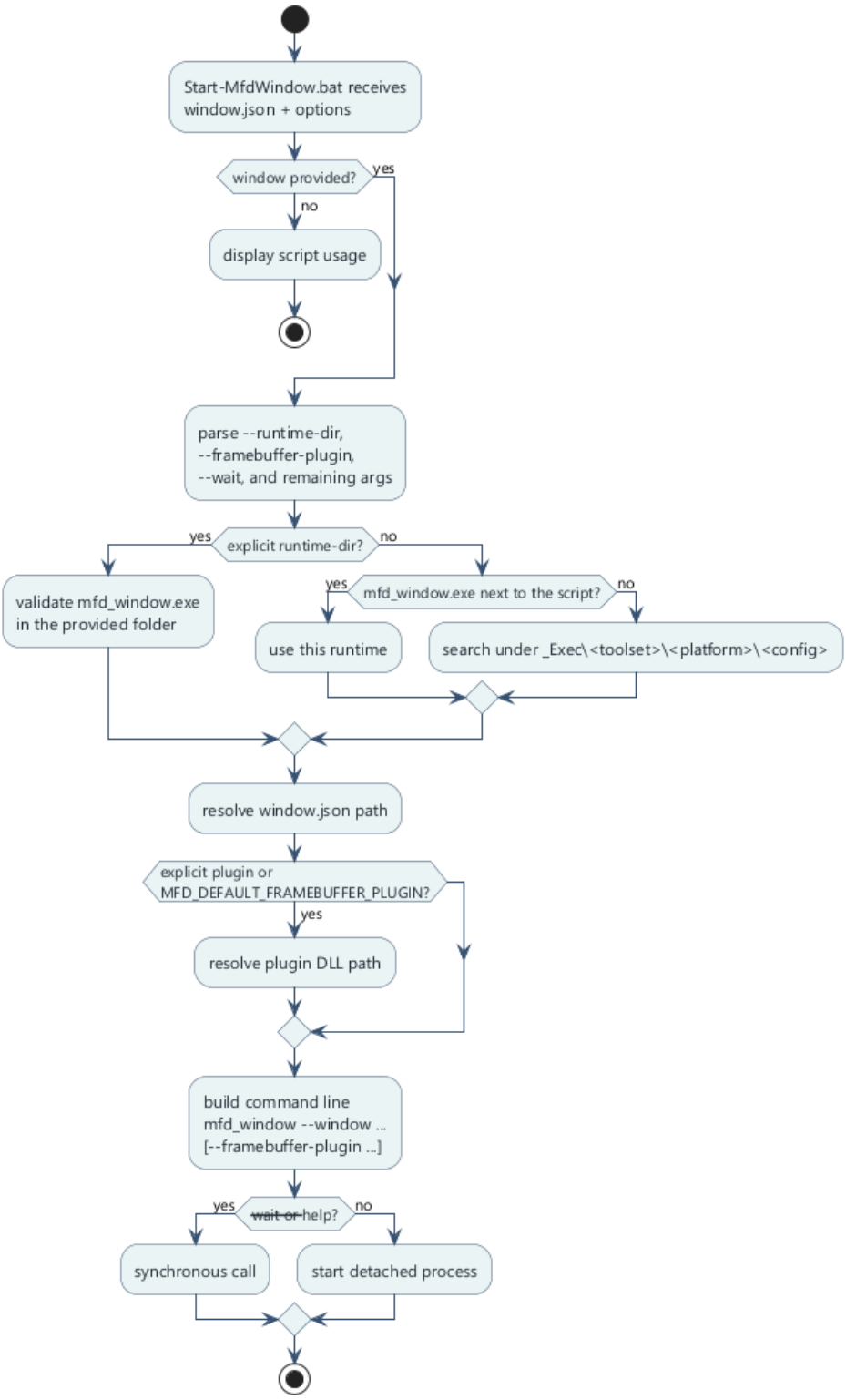
- compile as a DLL
- include `mfd/window/WindowLauncherPlugin.h`
- export `MfdGetWindowFramebufferPluginApi`
- keep the DLL boundary `noexcept`
- keep implementation detail in private C++ types behind `plugin_context`

7.14 Final integration checklist

Before wiring the plugin into a full runtime launch, verify all of the following:

70. 1. the DLL exports `MfdGetWindowFramebufferPluginApi`
71. 2. `info.abi_version` matches `MFD_WINDOW_FRAMEBUFFER_PLUGIN_ABI_VERSION`
72. 3. `info.requested_pixel_format` is set to the layout your consumer expects
73. 4. `init`, `submit_frame`, `close`, and `destroy` are all populated
74. 5. `submit_frame` validates the descriptor before touching pixels
75. 6. pixels are copied or handed off before the callback returns
76. 7. the async path does not block the render thread
77. 8. the plugin can be exercised with a synthetic stack-allocated frame

8. Launch a window with a script and a framebuffer plugin



Launch script flow

The repository already ships a reusable batch launcher:

CODE TEXT

```
Scripts/Start-MfdWindow.bat
```

8.1 Use the launcher as the stable runtime entry point

Minimum call:

```
Start-MfdWindow.bat <window.json>
```

Full call shape:

```
Start-MfdWindow.bat <window.json> [--runtime-dir <dir>] [--framebuffer-plugin <plugin.dll>]
[--wait] [extra launcher args]
```

Use this script when you want one repeatable runtime entry point for authored windows, staged binaries, and optional framebuffer-plugin injection.

8.2 Supported arguments

Argument	Use
<window.json>	authored window to load
--runtime-dir <dir>	explicit folder that contains mfd_window.exe
--framebuffer-plugin <plugin.dll>	plugin DLL to inject into the runtime
--wait	keep the launcher attached to the child process
extra launcher args	forwarded to mfd_window

8.3 How the script resolves mfd_window.exe

Resolution order:

78. 1. the folder passed through `--runtime-dir`
79. 2. the script folder if `mfd_window.exe` is already there
80. 3. `_Exec/<toolset>/<platform>/<config>`
81. 4. a fallback scan under `_Exec`

This makes the same script usable from a staged delivery tree and from a developer build tree.

8.4 How plugin resolution works

The plugin path can come from:

- `--framebuffer-plugin`
- `MFD_DEFAULT_FRAMEBUFFER_PLUGIN`

The script then resolves the path relative to:

- the explicit absolute path if already absolute
- the runtime directory
- the repository root

- the script directory
- the current working directory

8.5 Example of a preconfigured project launcher

`Scripts/Start-MfdMinimal.bat` is intentionally small:

CODE BAT
```bat
@echo off
setlocal
set "MFD_DEFAULT_FRAMEBUFFER_PLUGIN=mfd_framebuffer_stdout_plugin.dll"
call "%~dp0Start-MfdWindow.bat" "assets/windows/demo_pages_minimal.json" %*
exit /b %ERRORLEVEL%
```

This is a good pattern for project-specific launchers: set one default window, optionally set one default plugin, then delegate everything else to `Start-MfdWindow.bat`.

`Scripts/Start-MfdPackageTest.bat` follows the same pattern for the packaged SDK validation window. It launches the lightweight `assets/windows/package_test_window.json` asset set, which contains one rounded circle inside one small runtime window.

8.6 Example with an explicit custom plugin

CODE POWERSHELL
```powershell
.\Scripts\Start-MfdWindow.bat `
  "assets/windows/demo_pages_minimal.json" `
  --framebuffer-plugin "build\vs2022-win32\examples\Debug\my_framebuffer_plugin.dll" `
  --wait
```

Equivalent direct runtime call:

CODE POWERSHELL
```powershell
mfd_window --window assets/windows/demo_pages_minimal.json --framebuffer-plugin
my_framebuffer_plugin.dll
```

8.7 Why the script is still useful when the CLI exists

The script already solves three tedious runtime problems:

- finding the staged executable
- resolving the plugin path from several likely locations
- keeping a consistent project-local launch convention

That is why it remains the preferred operator entry point even when the underlying `mfd_window` CLI is available.

8.8 Write your own project launcher

CODE BAT
```bat
@echo off
setlocal
set "MFD_DEFAULT_FRAMEBUFFER_PLUGIN=my_encoder_plugin.dll"
call "%~dp0Start-MfdWindow.bat" "assets/windows/my_window.json" %*
exit /b %ERRORLEVEL%
```

Keep the custom script small. Let the shared launcher own path resolution and argument parsing.

8.9 Launch checklist with a framebuffer plugin

Before launch, verify:

- the window JSON exists
- `mfd_window.exe` exists in one resolvable location
- the plugin DLL exists in one resolvable location
- the plugin exports `MfdGetWindowFramebufferPluginApi`
- the plugin ABI version matches the runtime

8.10 If the plugin does not load

Check in this order:

82. 1. incorrect DLL path
83. 2. missing DLL runtime dependencies
84. 3. missing exported entry point
85. 4. incomplete callback table
86. 5. invalid `struct_size` or `abi_version`
87. 6. crash or invalid return inside `init`

9. Overall recommended checklist

9.1 End-to-end project workflow

1. author the window and its pages in `mfd_editor`
2. save the authored assets in `assets/`
3. expose only the runtime surface the client really needs
4. declare dynamic template bindings and page strobe behavior intentionally
5. regenerate `Ui.h`, `Ui.cpp`, and `<window>.generated.map`
6. rebuild the client target that includes the generated `.cpp`
7. choose the runtime host path: `mfd_window` or one application embedding `mfd_runtime_api`
8. publish coherent batches through `CommandClient` or `LatestBatchPublisher`
9. poll feedback if page activity or strobe capture matters
10. attach a framebuffer plugin only when the `mfd_window` host must export the rendered image

9.2 Design errors to avoid

- author directly in `_Exec`
- expose too many primitives without operational value
- keep stale generated outputs after an authored contract change
- scatter `SendBatch()` calls throughout the frame
- bypass runtime feedback when logic depends on authoritative page or capture state
- expose `mfd_api` directly to one embedding application when `mfd_runtime_api` is the intended public boundary
- keep framebuffer pointers after `submit_frame` returns

9.3 Files every integrator should know

File	Why it matters
assets/windows/<window>.json	generator and runtime entry point
assets/windows/<window>.generated.map	transport ids and compatibility hash
mfd_client_api/generator/scripts/generate_ui.py	generation logic and --print-inputs behavior
mfd_client_api/generator/ClientApiGenerator.cmake	official CMake integration path
mfd_client_api/include/mfd/client/ClientSdk.h	umbrella standalone SDK header for loader, transport, feedback, and publisher helpers

mfd_client_api/include/mfd/client/GeneratedUiSupport.h	single generated-code include bridging typed UI code to the packaged client SDK
mfd_common_api/include/mfd/control/CommandClient.h	low-level typed command sender, republished through ClientSdk.h in the packaged client SDK
mfd_api/include/mfd/io/JsonLoader.h	low-level authored window and page loader; this is the only header republished from the low-level API into the packaged client SDK through ClientSdk.h
mfd_client_api/include/mfd/client/Animation.h	generated-runtime support header used by GeneratedUiSupport.h ; not the normal user include for window-specific code
mfd_client_api/include/mfd/client/LatestBatchPublisher.h	latest-state asynchronous publisher
mfd_runtime_api/include/mfd/runtime_api/RuntimeSession.h	public offscreen runtime session entry point
mfd_runtime_api/include/mfd/runtime_api/OffscreenSurface.h	public resizable offscreen output surface
mfd_window_plugin_api/include/mfd/window/WindowLauncherPlugin.h	stable framebuffer-plugin ABI
Scripts/Start-MfdWindow.bat	runtime and plugin launch path
Scripts/Start-MfdPackageTest.bat	lightweight launcher for the package-

	consumer validation window
examples/offscreen_viewer/src/main.cpp	reference embedding example for the offscreen runtime package
examples/client_test_package/CMakeLists.txt	reference find_package(MFDStudioClientApi) consumer using the staged delivery
_Deliveries/packages_windows/MFDStudioRuntimeApi/build/native/cmake/MFDStudioRuntimeApiConfig.cmake	packaged development SDK entry point for offscreen embedding
_Deliveries/packages_windows/MFDStudioClientApi/build/native/cmake/MFDStudioClientApiConfig.cmake	packaged development SDK entry point exported for external CMake clients
_Deliveries/packages_bin_windows/MFDStudioClientApi.Install/_Exec/<toolset>/<platform>/<config>/mfd_client_api.dll	standalone runtime payload required by packaged generated clients

9.4 Final takeaway

The clean MFDStudio path is:

- author the visual structure in the editor
- generate one typed C++ contract from the authored window
- publish one coherent client batch per frame
- let runtime feedback confirm page activity and strobe capture
- choose `mfd_runtime_api` for direct embedding and keep framebuffer plugins for the `mfd_window` path
- copy framebuffer pixels safely before crossing into async plugin logic when a plugin is involved

This keeps the repository modular, the client surface intentional, and the runtime behavior debuggable.

Appendix A. Quick Reference

This appendix is the one-page cheat sheet for the API calls that appear most often in a production client.

API call	Signature	3-word use
activate page	bool ActivatePage(const GeneratedPage& page)	select active page
set page view	bool SetPageView(const GeneratedPage& page, Vec2 center, float zoom)	pan and zoom
access whole window	WindowDisplay& Window() noexcept	window display state
start local cycle	void Run() noexcept	drop staged dirt without consuming a pending initialize
authored reset	void Initialize() noexcept	authored runtime reset
collect dirty commands	std::vector<mfd::UserCommand> BuildBatch()	gather dirty commands
build reset commands	std::vector<mfd::UserCommand> BuildResetBatch()	standalone reset batch
build tracked batch	mfd::CommandBatch BuildCommandBatch(std::uint32_t sequence = 0)	attach hash sequence
build tracked reset	mfd::CommandBatch BuildResetCommandBatch(std::uint32_t sequence = 0)	tracked reset batch
send one frame	bool SendBatch(const mfd::CommandBatch& batch)	publish current frame
queue freshest frame	bool SubmitLatest(LatestBatchPublisher&, std::uint32_t sequence = 0)	stream latest state
queue reset batch	bool SubmitReset(LatestBatchPublisher&, std::uint32_t sequence = 0)	stream authored reset
poll runtime feedback	std::size_t PollFeedback(IExchangeChannel&, std::size_t maxMessages = 64,	drain feedback packets

	std::string* error = nullptr)	
check page activity	bool IsActive() const noexcept	runtime page active
check dynamic validity	bool IsAlive() const noexcept	stale handle guard
check capture state	bool IsStrobeCaptured() const noexcept	runtime target captured on the active page strobe
opt one dynamic set into strobe magnetization	void SetStrobeMagnetEnabled(bool)	allow this template family to snap/follow the strobe while keeping capture feedback independent